



Solar Analytics

Analysis of ISEM Solar Trading

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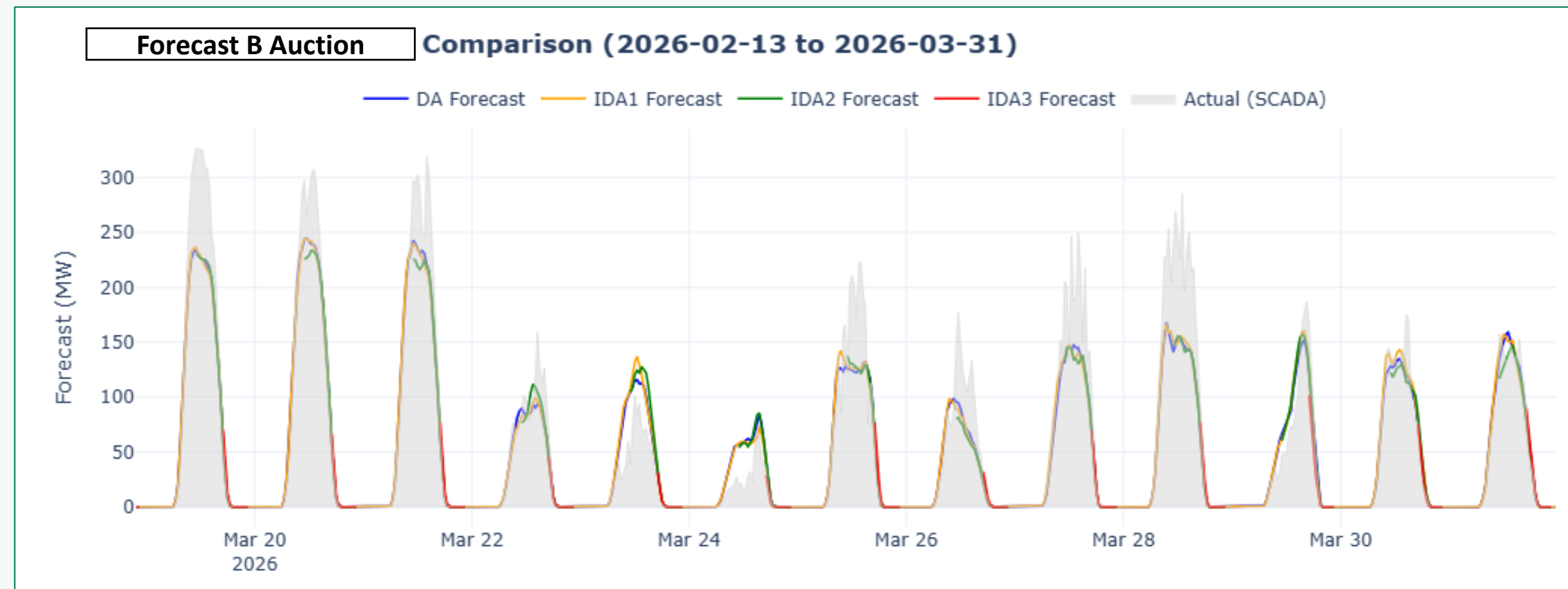
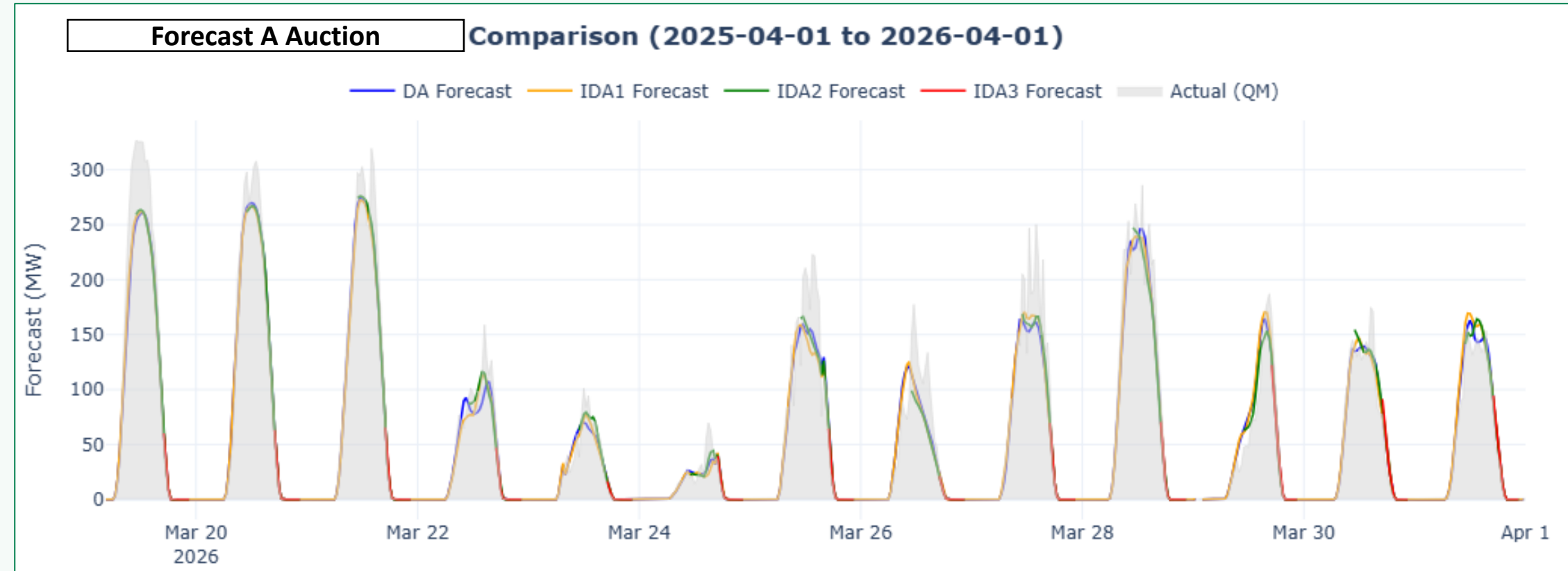
Objective:

- The overall objective of this project is to investigate how we could trade our solar positions better across different auctions.
- Currently, we have a tendency to just sell most of our solar positions at the day-ahead auction and either modify positions at later auctions or just settle our positions in the balancing market.
- This analysis investigates if this is the best strategy, and if there are other more strategic ways to manage our positions in order to maximise our revenue.
- Analysis and strategies were back tested across the last 12 months to see if we could have done things differently – and give a framework for how we could possibly change how we do things going forward.
- The first section of the slides focuses on forecasting accuracy analyses, followed by strategy.



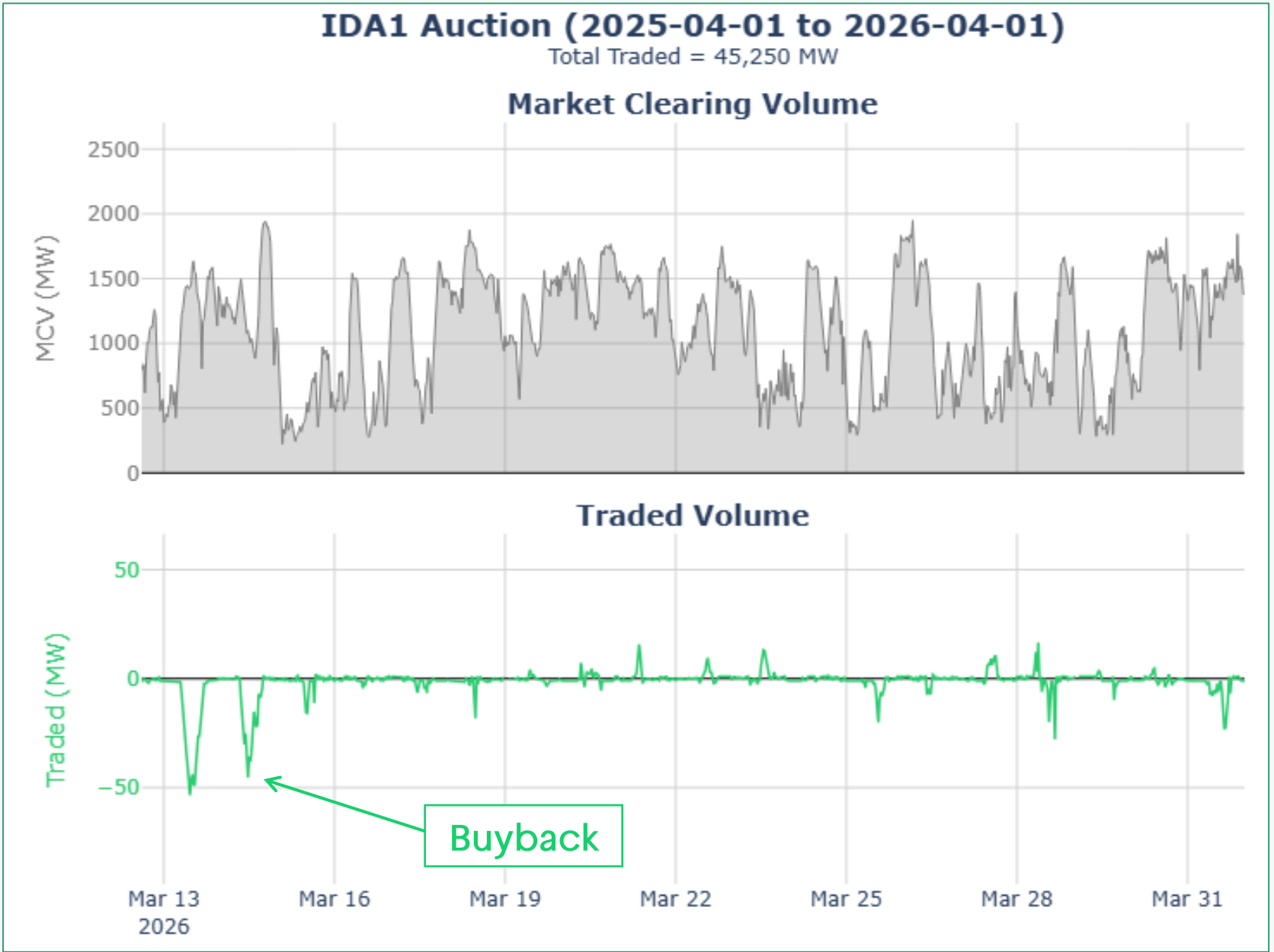
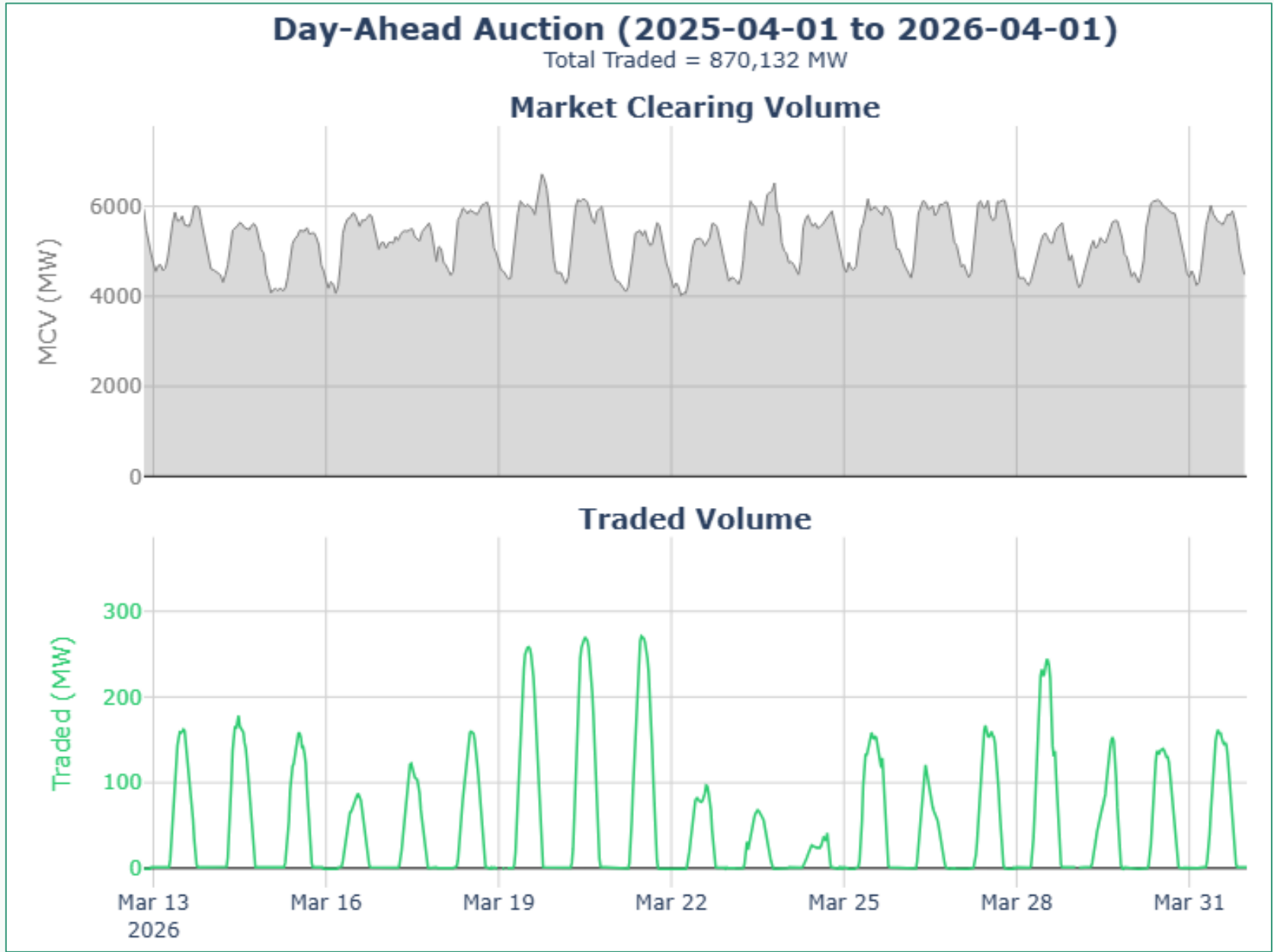
Solar Forecast Accuracy

- The following slides look at the performance and comparison of the two forecast providers, Forecast A vs Forecast B.
- Dispatch curtailment and constraint volumes were ignored as the generation forecasts don't take dispatch down into consideration.
- Some of the following key indicators are used to quantify forecast accuracy:
 - ME/MBE = mean error/bias
 - MAE = mean absolute error
 - RMSE = root mean squared error
 - MAPE = mean absolute percentage error
- **Error Sign Convention:**
- **Error = Forecast – Metered Generation**
Negative = Long/Under-forecasting
Positive = Short/Over-forecasting



Current Auction Trading Volumes

Auction	MCV (MW)	Traded (MW)	Sold (MW)	Bought (MW)	% of MCV	Buyback %
Day-Ahead	44,459,418.4	870,131.6	867,190.8	2,940.8	1.96	0.34
IDA1	20,077,865.5	45,250.5	14,503.0	30,747.5	0.23	67.95
IDA2	2,225,002.5	27,731.1	10,083.2	17,647.9	1.25	63.64
IDA3	515,106.0	5,762.9	1,995.8	3,767.1	1.12	65.37



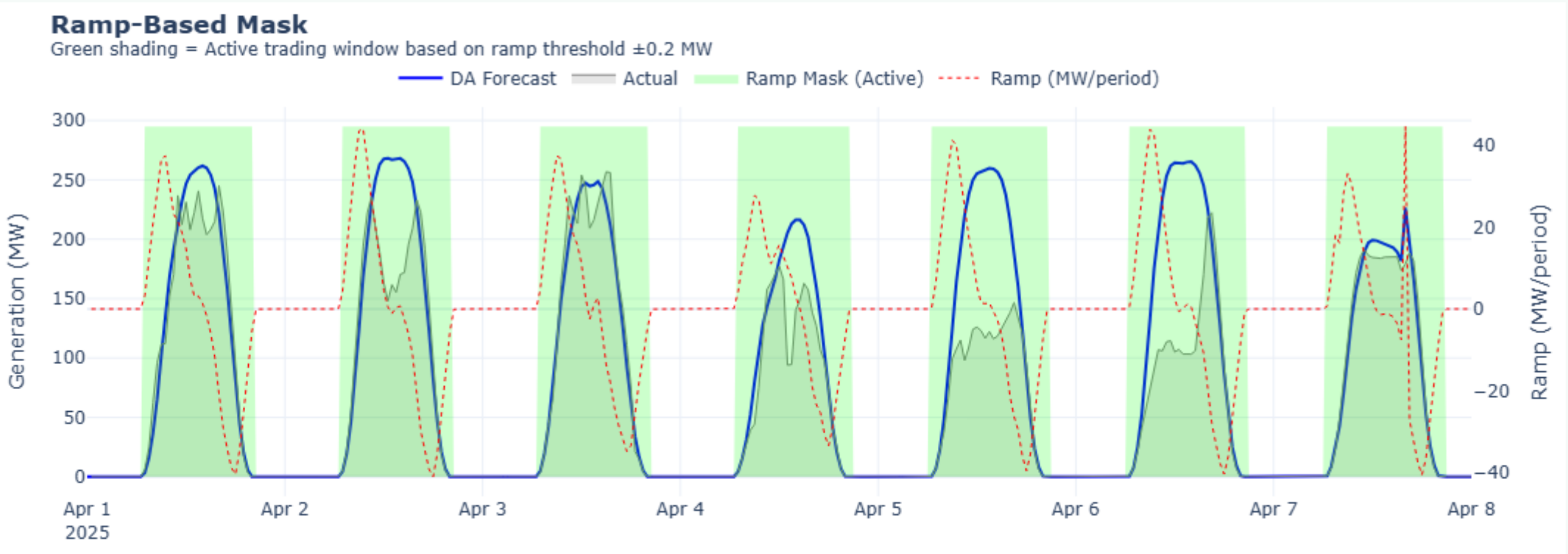
Filtering the Forecasts

- At first, all periods were being analysed (including night), which caused in the error statistics to be inaccurate.
- To improve accuracy, a static mask was used to try and filter out the periods of no solar generation.
- However, due to the solar path changing throughout the year, the periods where solar generation equals 0 MW varies.
- In order to handle this, a dynamic mask approach was used.
- The solar ramp was defined as being the difference in generation between two consecutive 30-min periods (or the derivative of the solar curve).
- A ramp threshold was set to 0.2 MW.
- This meant that only periods where solar generation was occurring were to be analysed, ultimately yielding more accurate results.

Auction:	Static Mask Definition:
DA & IDA1 Mask:	06:00 – 23:00
IDA2 Mask	11:00 – 23:00
IDA3 Mask	17:00 – 23:00

$$\text{Ramp} = \frac{d}{dt}(\text{Solar Curve})$$

Auction:	Dynamic Mask Definition:
DA & IDA1 Mask:	[0.2 MW < Ramp < 0.2 MW]
IDA2 Mask	11:00 – [Ramp < 0.2 MW]
IDA3 Mask	17:00 – [Ramp < 0.2 MW]



Forecast A Accuracy

- The expected outcome for the error distribution would logically be DA to IDA3, least to most accurate as we get closer to delivery.
- However this was not the case, as seen in the *Error (All Hours)* table. The error data was being flattened by the 0 and NaN periods within the data outside of the solar hours.
- When the static mask was applied, the distribution became more aligned with what was expected.
- When the dynamic mask was applied, the results became even more accurate. This led to the adjusted error results being much more in-line with expectation.
- A positive mean error across DA, IDA1, and IDA2 suggest we are usually long in these auctions, compared to being slightly short in IDA3.
- IDA3 was by far the most accurate forecast, despite having a high MAPE (likely caused by small division, yielding high numbers).
- IDA2 had the highest MAE and RMSE, likely due to it overcompensating for volatility.

Error (All Hours)

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	2.74	10.82	22.97	71.1%
IDA1	2.34	10.6	22.65	69.7%
IDA2	2.83	15.5	27.55	62.0%
IDA3	-0.42	4	10.08	96.2%

Adjusted Error (Static Mask)

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	3.86	15.25	27.29	63.6%
IDA1	3.31	14.95	26.91	61.7%
IDA2	2.83	15.50	27.55	62.0%
IDA3	-0.02	3.10	8.20	99.0%

Adjusted Error (Dynamic Mask)

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	5.07	19.95	31.21	57.5%
IDA1	4.34	19.55	30.77	55.4%
IDA2	3.85	21.08	32.13	53.5%
IDA3	-0.88	8.47	14.68	93.4%

Forecast B Accuracy

- Forecast B is trained on historic SCADA, and has been in use since 13/02/2026.
- A negative bias can be seen in the mean error, indicating that Forecast B systematically under-forecasts generation.
- This is likely causing us to have a long position.
- IDA2, despite being closer to delivery than DA and IDA1, has the highest error.
 - This is likely once again due to IDA2 trying to forecast the volatility caused by varying cloud coverage.

Error (All Hours): 13/02/2026 – 26/04/2026

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	-2.11	12.80	26.55	79.5%
IDA1	-2.17	12.51	26.11	78.4%
IDA2	-6.83	18.63	32.48	70.1%
IDA3	-0.80	3.35	9.36	117.5%

Adjusted Error (Dynamic Mask): 13/02/2026 – 26/04/2026

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	-3.77	22.96	35.58	62.2%
IDA1	-3.87	22.44	34.99	60.6%
IDA2	-9.18	25.04	37.66	59.8%
IDA3	-1.60	6.84	13.41	136.0%

Forecast Accuracy Comparison

Key Conclusions:

- In order to make a fair comparison, both Forecast A and Forecast B were analysed from 13/02/2026 to 26/04/2026.
- Both forecasts followed the same pattern – with IDA2 forecasting being the least accurate.
- Forecast B had a lower mean error overall – however this was likely due to errors cancelling out.
- Forecast B proved more accurate than Forecast A in IDA1 and IDA3.
- Comparing the MAE and RMSE, it suggests Forecast A's error is driven by error volatility rather than consistency.
 - E.g. For IDA2, Forecast A had a lower MAE than Forecast B, but it had a higher RMSE.

Forecast A (Dynamic Mask): 13/02/2026 – 26/04/2026

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	-6.89	22.36	35.01	36.3%
IDA1	-9.01	23.05	36.32	35.7%
IDA2	-11.71	24.91	38.17	35.0%
0.26	-5.18	8.43	18.84	54.5%

Forecast B (Dynamic Mask): 13/02/2026 – 26/04/2026

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	-3.77	22.96	35.58	62.2%
IDA1	-3.87	22.44	34.99	60.6%
IDA2	-9.18	25.04	37.66	59.8%
IDA3	-1.60	6.84	13.41	136.0%

Most Accurate Forecast?

Forecast	ME (MW)	MAE (MW)	RMSE (MW)	MAPE
DA	Forecast B	Forecast A	Forecast A	Forecast A
IDA1	Forecast B	Forecast B	Forecast B	Forecast A
IDA2	Forecast B	Forecast A	Forecast B	Forecast A
IDA3	Forecast B	Forecast B	Forecast B	Forecast A

Forecast A Forecast Accuracy

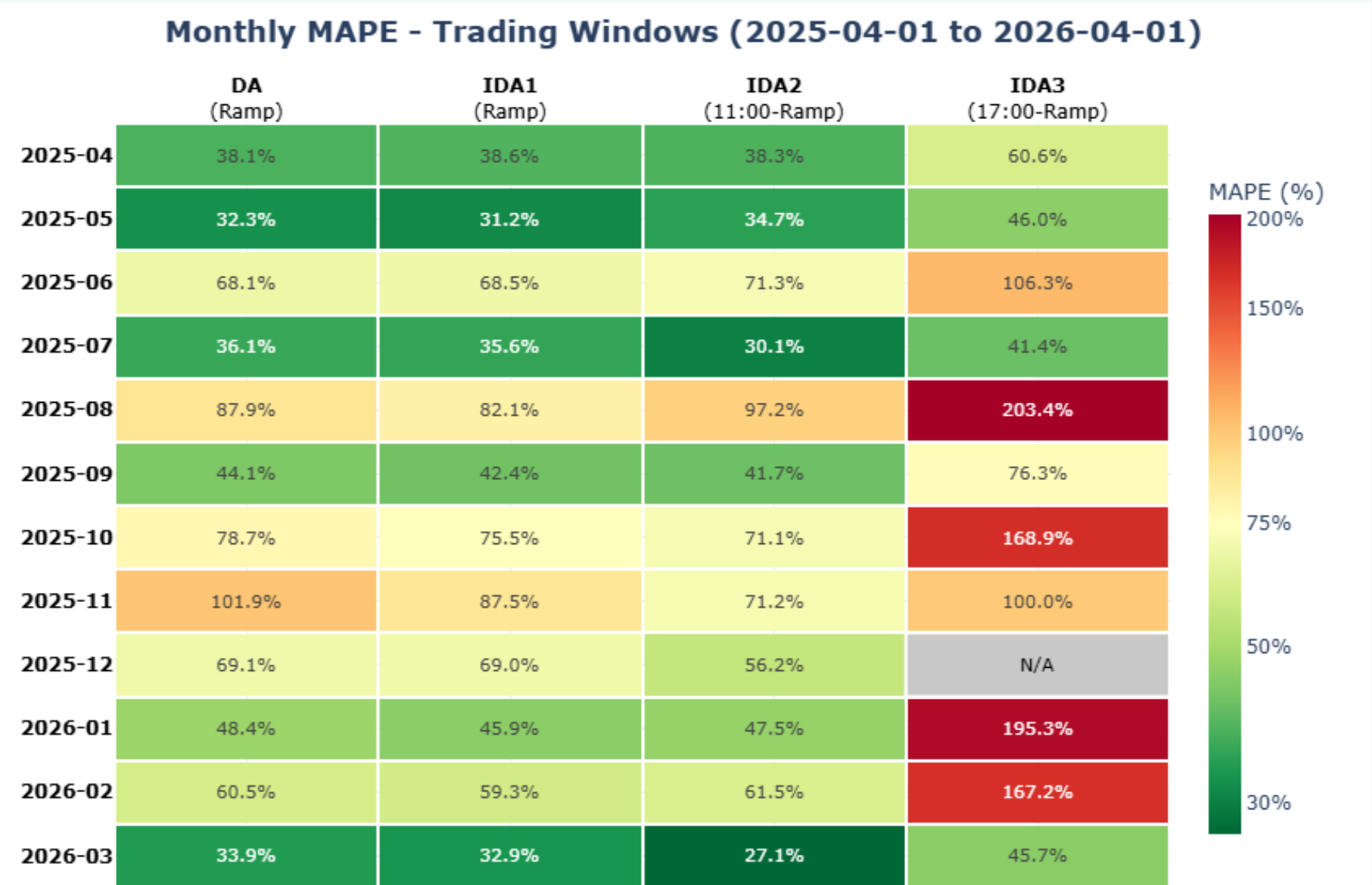
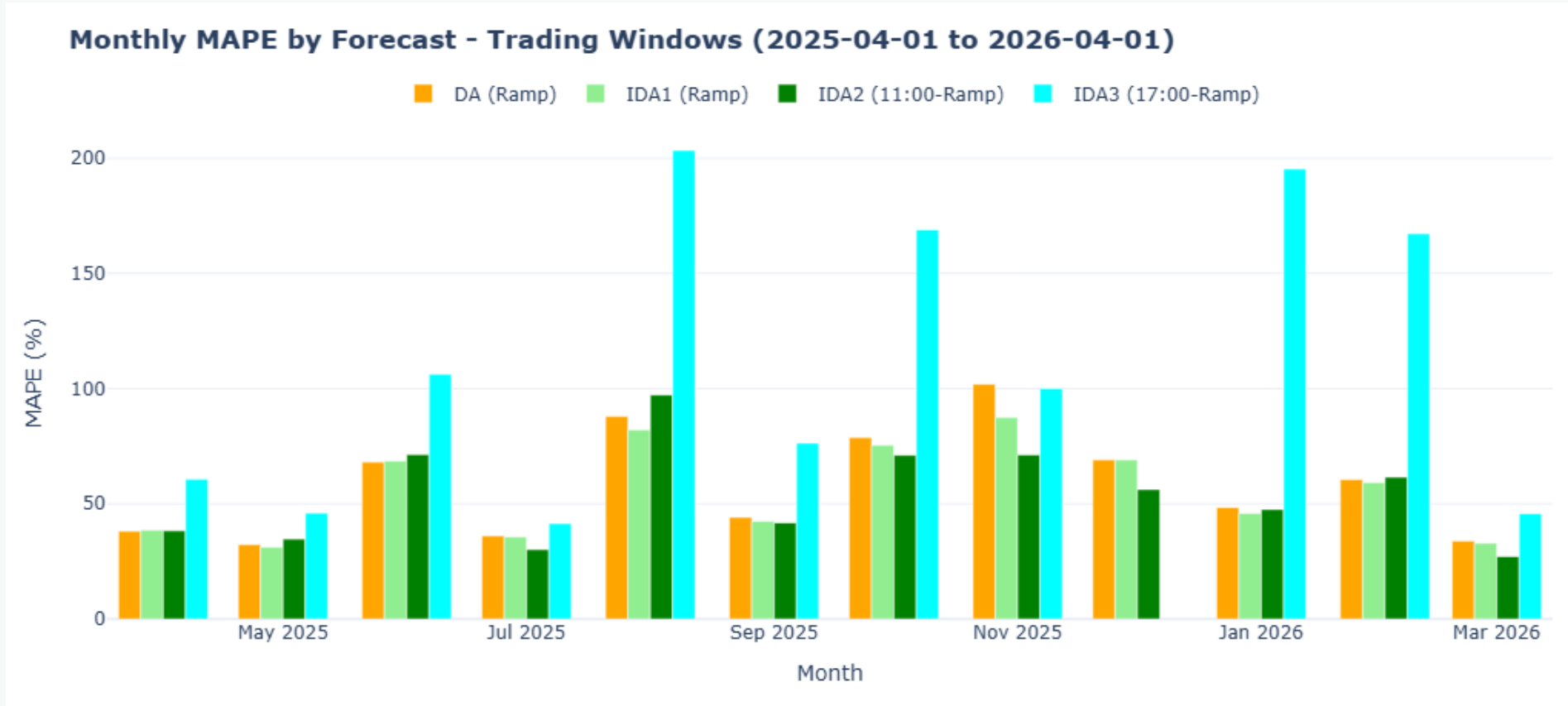
Looking at monthly MAPE levels, a few observations can be made.

IDA3 is Significantly Worse:

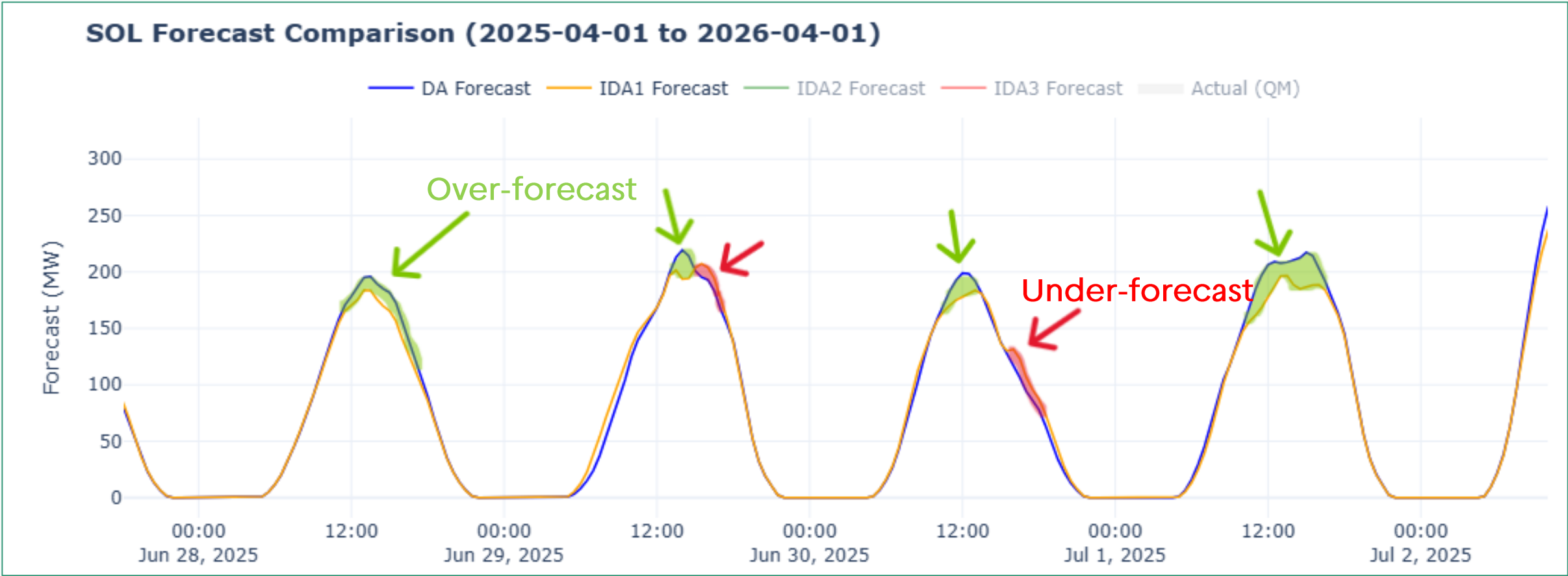
- August, October, January, and February are extreme outliers.
- MAPE for IDA3 is constantly higher than DA/IDA1/IDA2.
- However, this is likely due to IDA3 catching the ramp down, which results in MAPE yielded from dividing by small figures.
- December 2025 saw 0 MW of solar generated during IDA3, so it was omitted from the analysis.

Summer Performance Collapse:

- The summer months saw higher levels of solar generation, and hence also higher levels of variability.
- On days with high GHI levels, sudden cloud coverage/movement can trigger large forecast error swings.
- The extreme error in August was likely driven by a few individual days.



Forecast A Forecast Retracement



- **Over-forecast:** Latest forecast decreases from previous forecast.
 - $FCAST_t < FCAST_{t-1}$
- **Under-forecast:** Latest forecast increases from previous forecast.
 - $FCAST_t > FCAST_{t-1}$
- The below results are reflective of the degree of error we have seen previously.
- The largest amount of error is seen between IDA1 and IDA2.
- Overall all of our forecasts tend to over-forecast, leading to retracement.
- This highlights the systematic short risk bias.

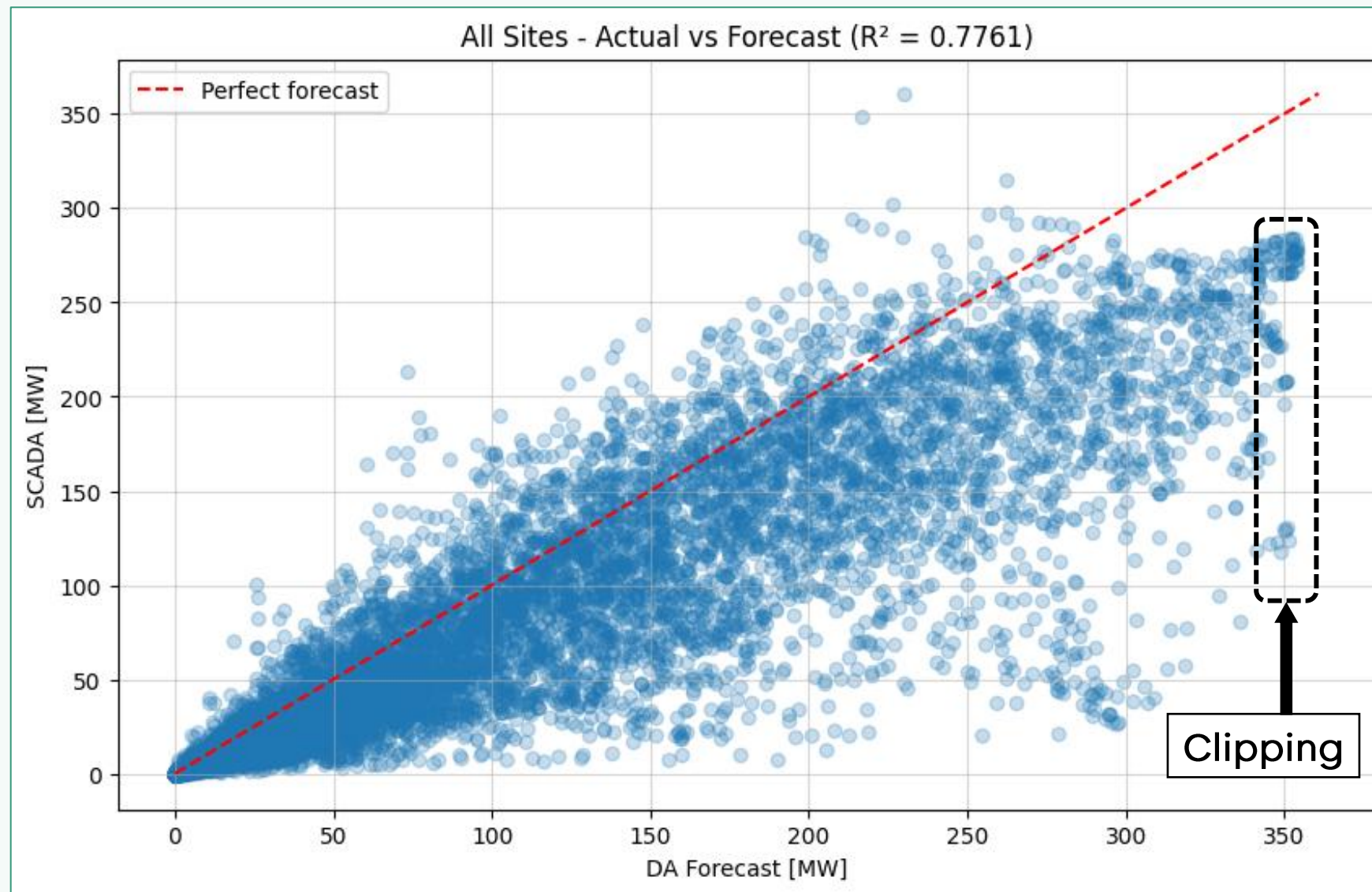
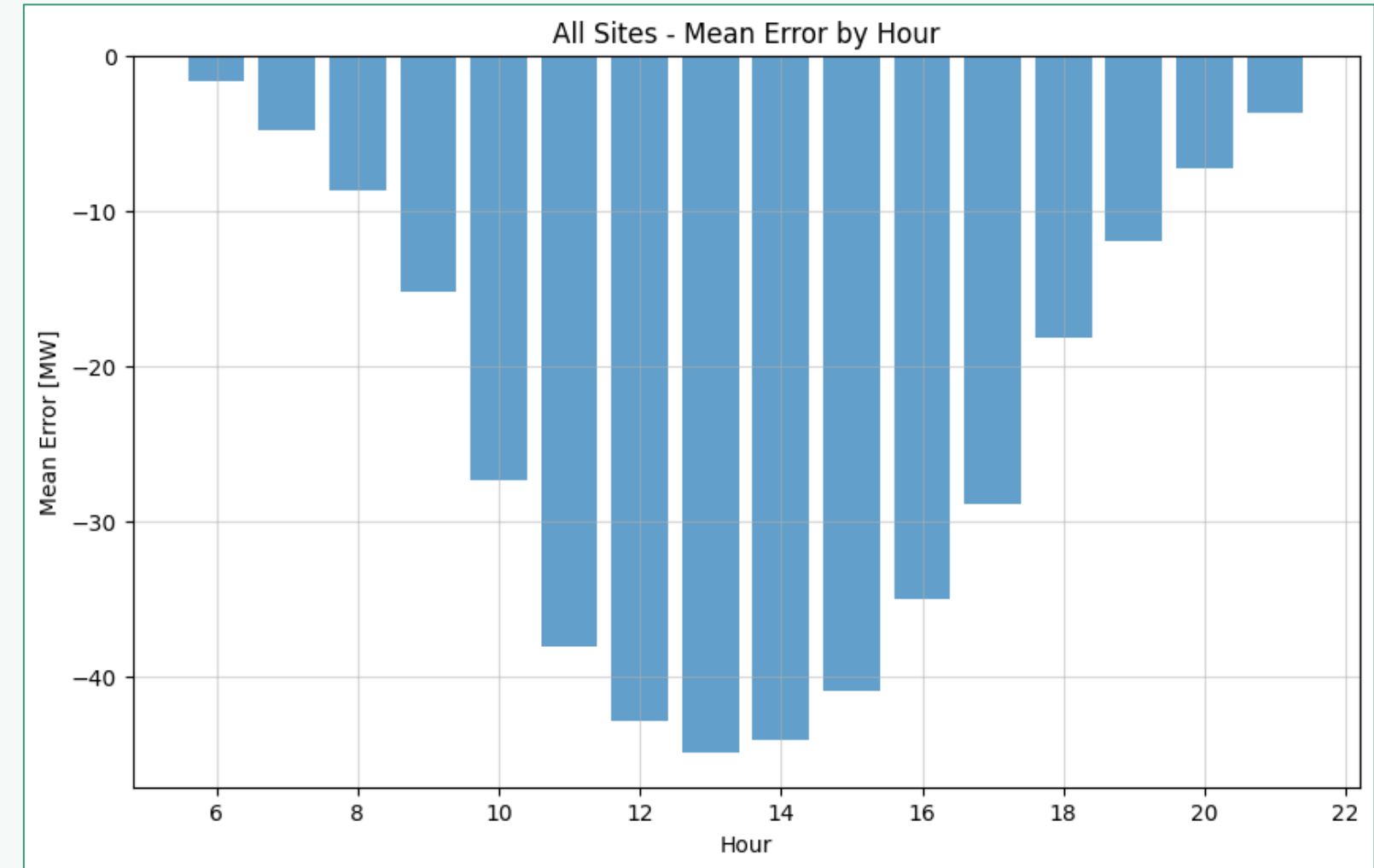
Over-forecast	Count	% Periods	Avg Magnitude (MW)
IDA1 < DA	4874	51.6%	5.39 MW
IDA2 < IDA1	3294	32.9%	7.93 MW
IDA3 < IDA2	1288	20.0%	6.02 MW

Under-Forecast	Count	% Periods	Avg Magnitude (MW)
IDA1 > DA	4281	45.8%	4.53 MW
IDA2 > IDA1	2818	29.9%	6.41 MW
IDA3 > IDA2	516	8.0%	2.08 MW

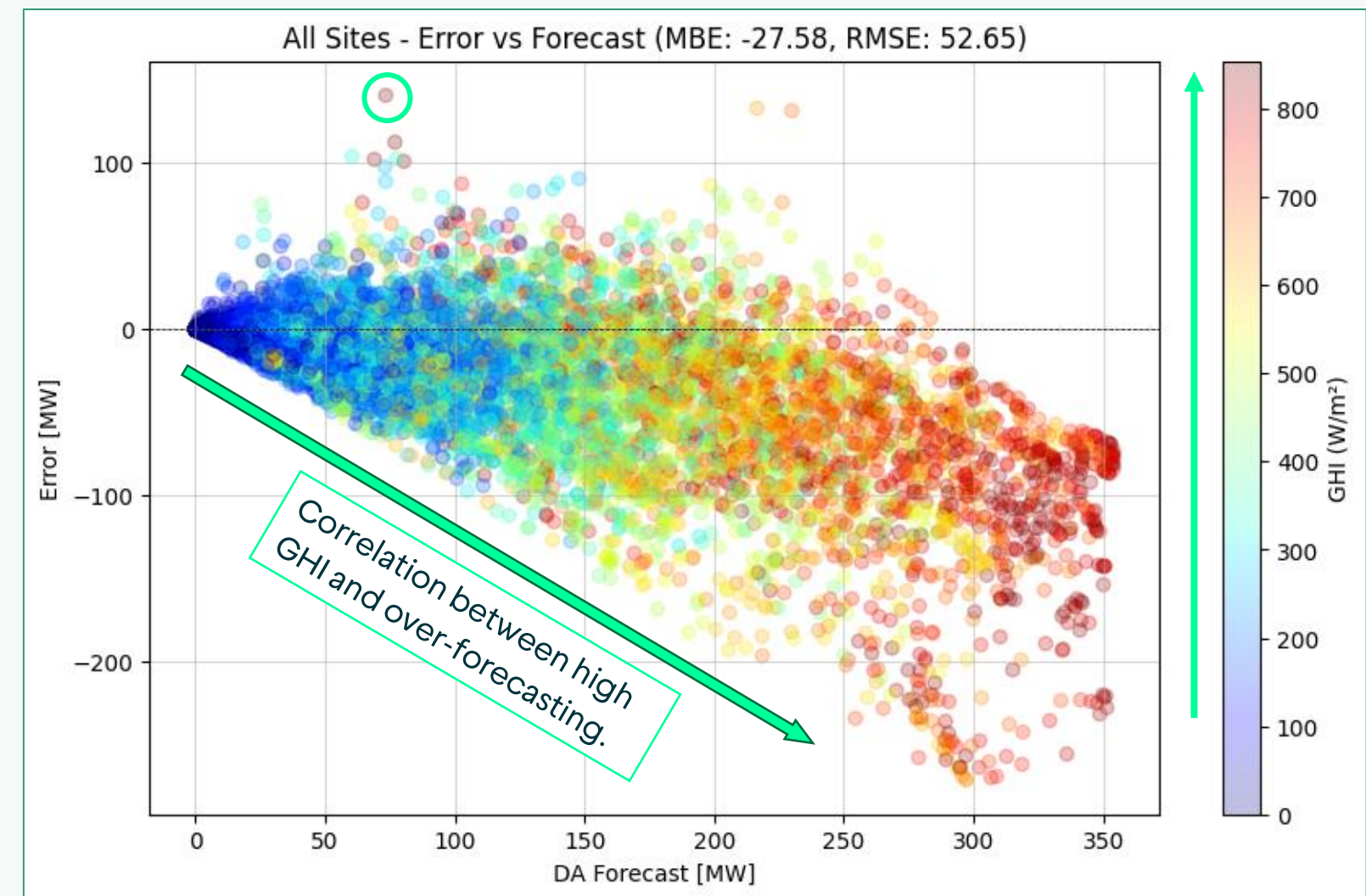
Forecast A Site Forecast Accuracy

- This section of analysis looks at the SCADA / DA forecast relationship at each solar site in our portfolio.
- Python's pvlib library was used to model clear sky GHI output at an aggregated coordinate of all our sites.

- A mask was applied to try and filter out inconsistencies:
06:30 – 22:00, excluding any periods where either SCADA, forecast, or both are NaN/0.



- It is worth noting that the portfolio's maximum export capacity (MEC) is 356.4 MW, which leads to a clipping effect as outlined in the plot above.



Forecast A Site Forecast Accuracy

'n' represents the amount of samples that were used in the analysis, with the red cells indicating significant data exclusion (mostly due to consistent SCADA outages).

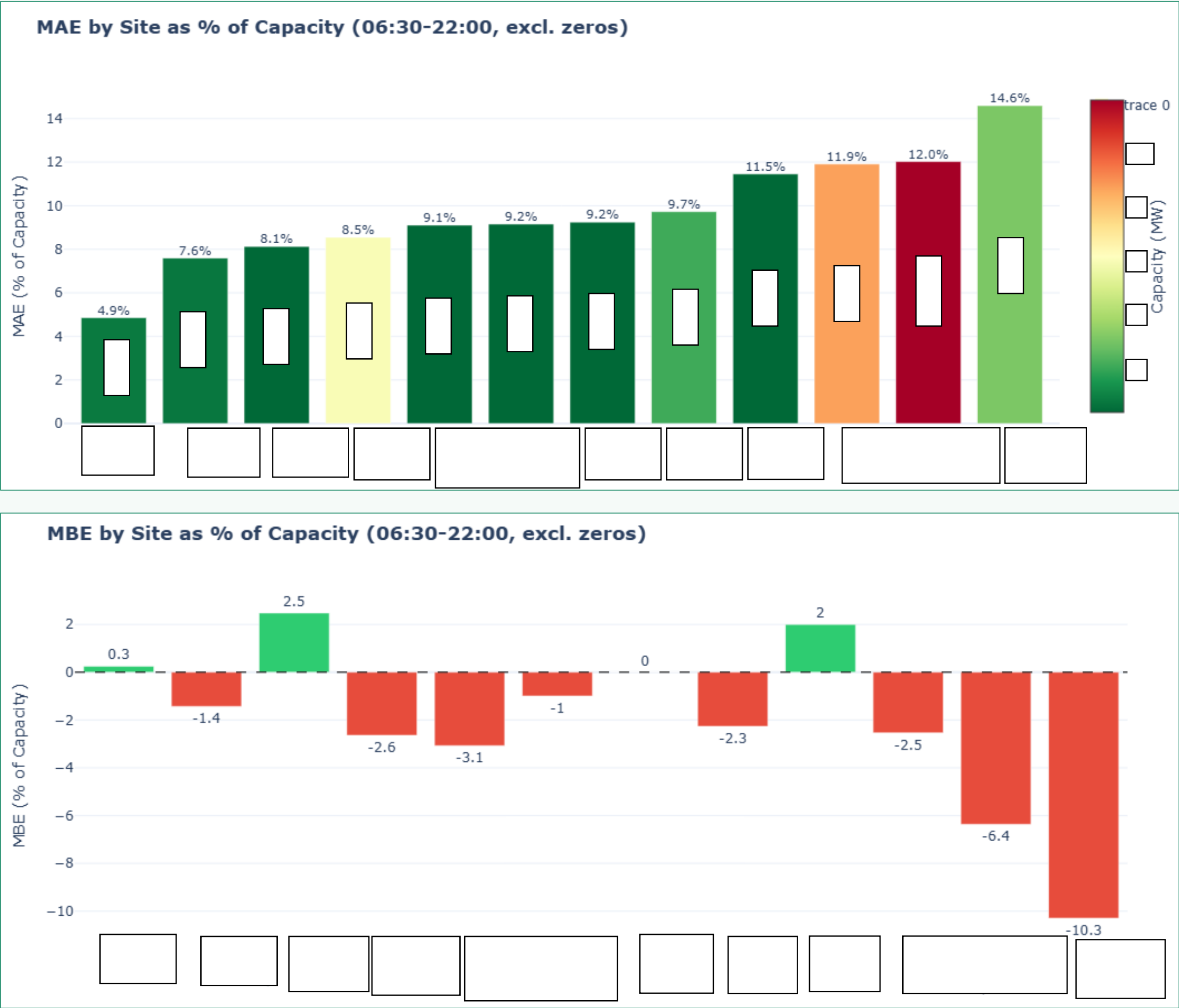
Portfolio Metric	Value	%
ME	-27.58 MW	-7.64%
MAE	33.53 MW	9.28%
RMSE	52.65 MW	14.58%
R ²	0.7761	-
Over-Forecast	-	80.4%
Under-Forecast	-	19.6%

Per-Site Forecast Error Statistics (sorted by nMAE)

Site	Capacity (MW)	n	ME (MW)	MAE (MW)	RMSE (MW)	nME (%)	nMAE (%)	nRMSE (%)	R ²	Over (%)	Under (%)
		3029	0.02	0.44	0.72	0.25	4.86	8.03	0.68	49.72	50.28
		8414	-0.12	0.63	0.95	-1.43	7.6	11.53	0.74	56.23	43.77
		3289	0.12	0.4	0.61	2.47	8.13	12.24	0.82	44.6	55.4
		7873	-1.58	5.13	8.12	-2.64	8.55	13.53	0.77	59.75	40.25
		8496	-0.15	0.43	0.67	-3.07	9.11	14.12	0.71	59.52	40.35
		3294	-0.05	0.45	0.72	-0.99	9.16	14.45	0.71	55.49	44.47
		3296	0	0.46	0.69	0.01	9.25	13.85	0.72	52.12	47.88
		7450	-0.49	2.12	3.23	-2.26	9.73	14.86	0.68	56.27	43.73
		3307	0.1	0.57	0.88	2	11.46	17.85	0.53	44.84	55.16
		6321	-2.21	10.37	15.9	-2.54	11.91	18.28	0.66	56.97	42.97
		8262	-7.58	14.31	21.85	-6.37	12.02	18.36	0.65	64.43	35.57
		5498	-3.25	4.61	7.35	-10.29	14.59	23.25	0.38	71.12	28.88

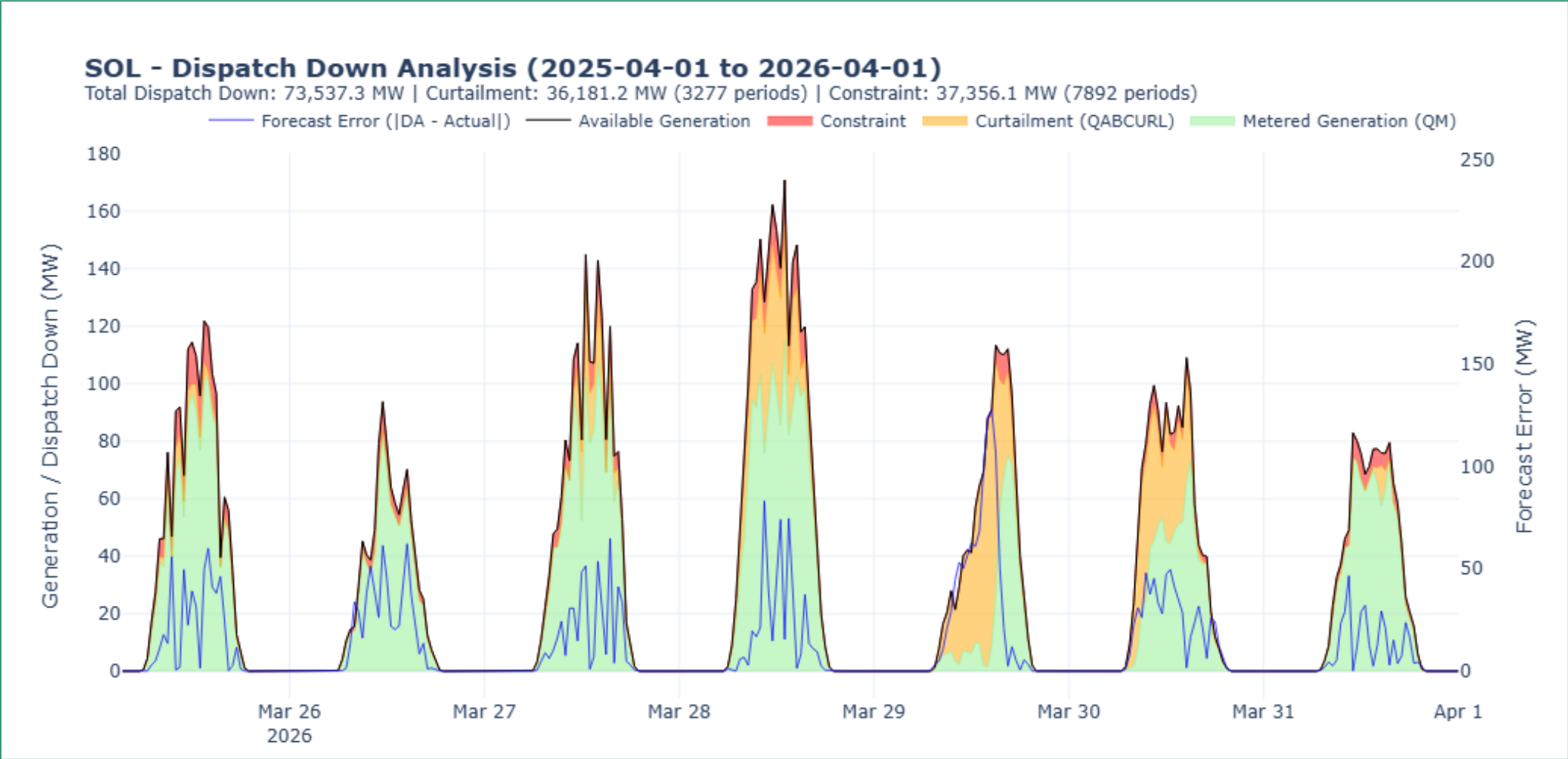
Forecast A Site Accuracy

- Looking at each Irish solar site individually, we can see how each one contributes to the overall forecast error.
- The site error distribution is skewed negatively with the larger sites positioned toward the upper tail. However there are some outliers such as [SITE NAME] which is a relatively large site with decent accuracy.
- For values where the MBE is negative, the SCADA is less than the forecasted volume (site over-forecasting). This means that we are likely to be short if trading to forecast.
- On average, the sites over-forecasted at all but two sites, meaning our solar book was more likely to be short across the last year. Therefore there appears to be a systematic short risk across our portfolio.
- It is worth mentioning that this analysis is based on SCADA, which excludes a lot of null periods. Further analysis will use metered generation.



Solar Dispatch Down Analysis

Dispatch Down = Curtailment + Constraint



- A mask was applied to the data to analyse only periods from 06:30 to 22:00.
- A positive correlation ($R = 0.765$) was found between dispatch down and forecast error.
- On average, there was an 9.17 MW difference in forecast error between periods with dispatch down versus periods without.
- A next step for analysis would be to investigate which site has the greatest dispatch down volume and why.

Period Analysis (30-min periods)	
Metric	Value
Periods Included (30-min)	11315 / 17521 (64.6%)
Periods with Dispatch Down	6783 (59.9%)
- with Curtailment	1990 (17.6%)
- with Constraint	5699 (50.4%)
Max Curtailment (30-min period)	132.75 MW
Max Constraint (30-min period)	72.01 MW
Overall Impact	
Dispatch Down % of Available	16.38 %
- Curtailment %	8.05 %
- Constraint %	8.33 %
Forecast Error vs Dispatch Down	
Correlation (DD vs Forecast Error)	0.765
Average Forecast Error during DD	23.95 MW
Average Forecast Error without DD	14.78 MW
Difference	+ 9.17 MW

Auction Split Strategy

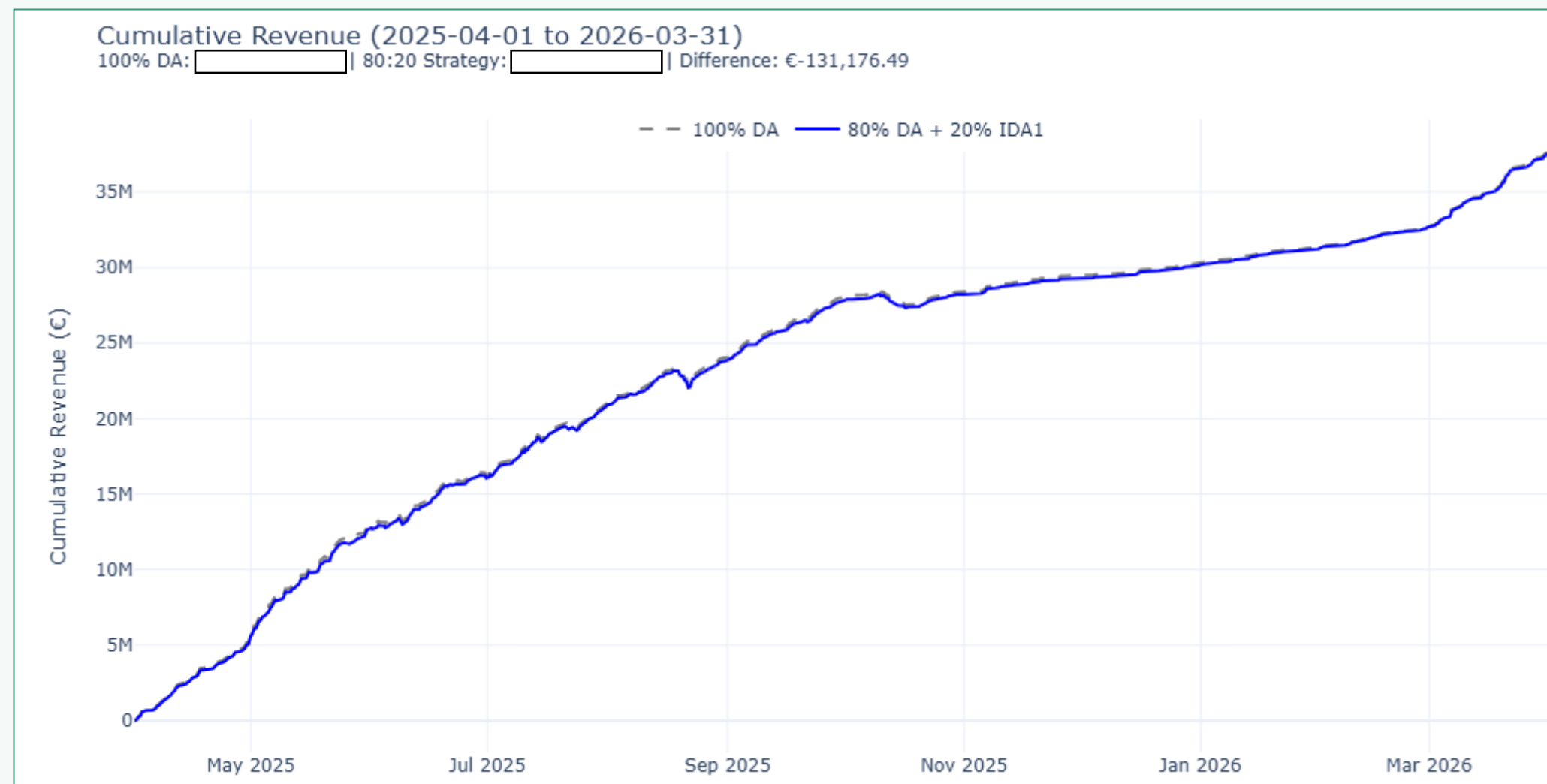
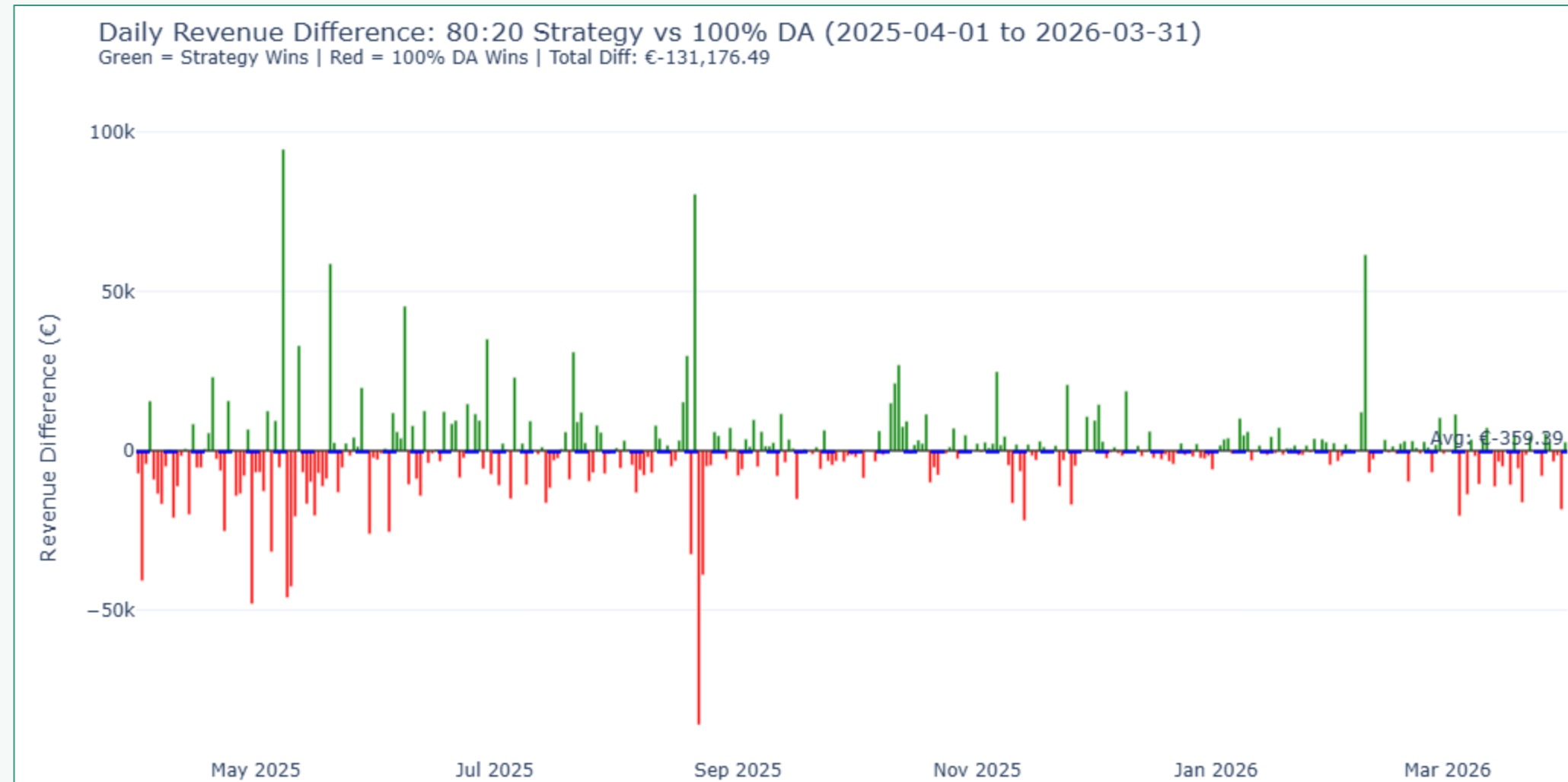
80% DA + Remaining % IDA1

- To test whether or not trading 100% to the DA forecast was the best strategy, a split was introduced.
 - 80% of the volume was traded to the DA forecast, approx. 20% to the IDA1 forecast, and the remainder was settled in the imbalance auction.
 - This was done to investigate whether we could capture more value in other auctions.

- Strategy results over a year:

Strategy	Value
80% DA + ~20% IDA1 Revenue	€XXX
100% DA Revenue	€XXX
Difference	- €131,176.49

- Trading to the DA forecast outperformed the split.

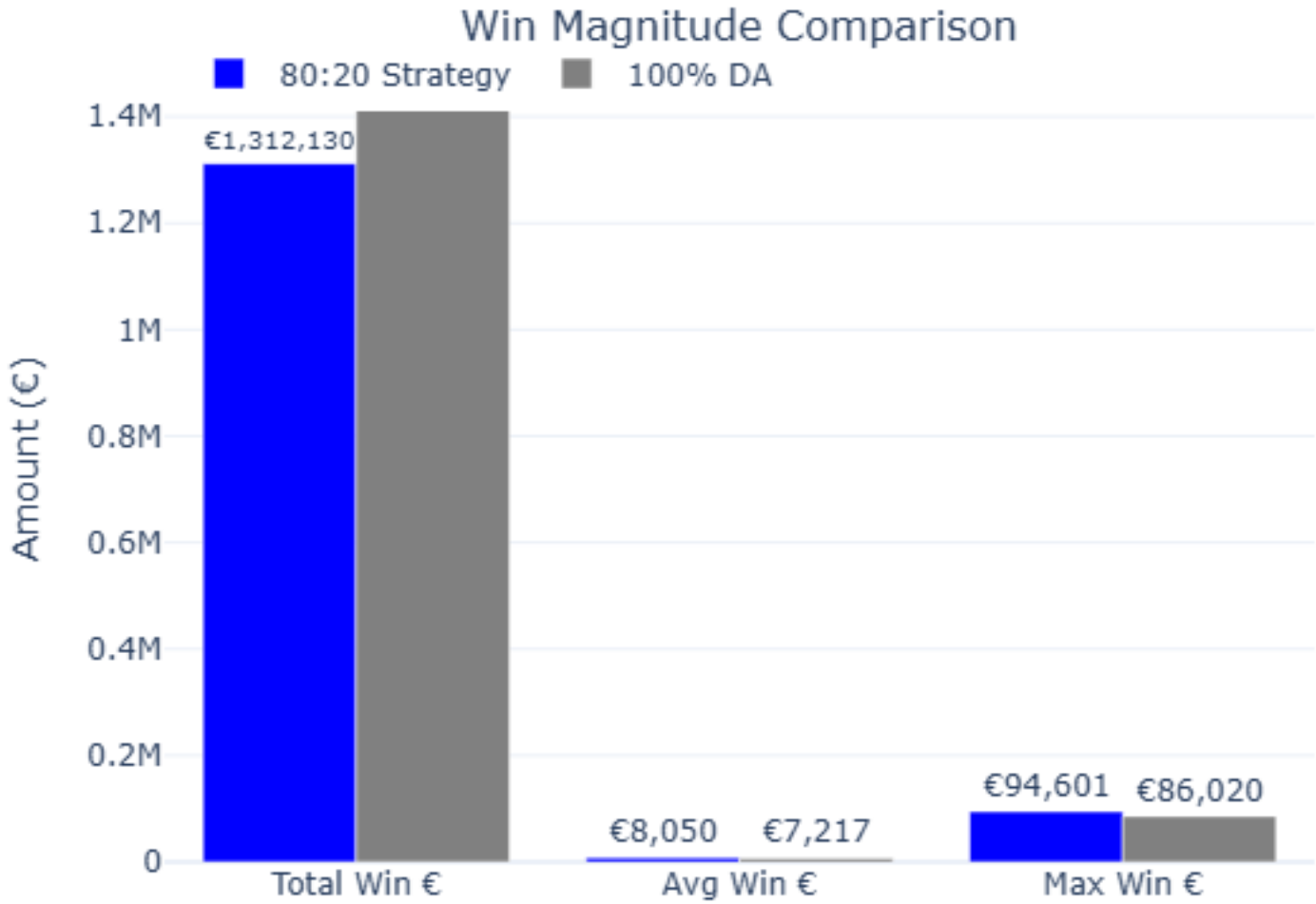
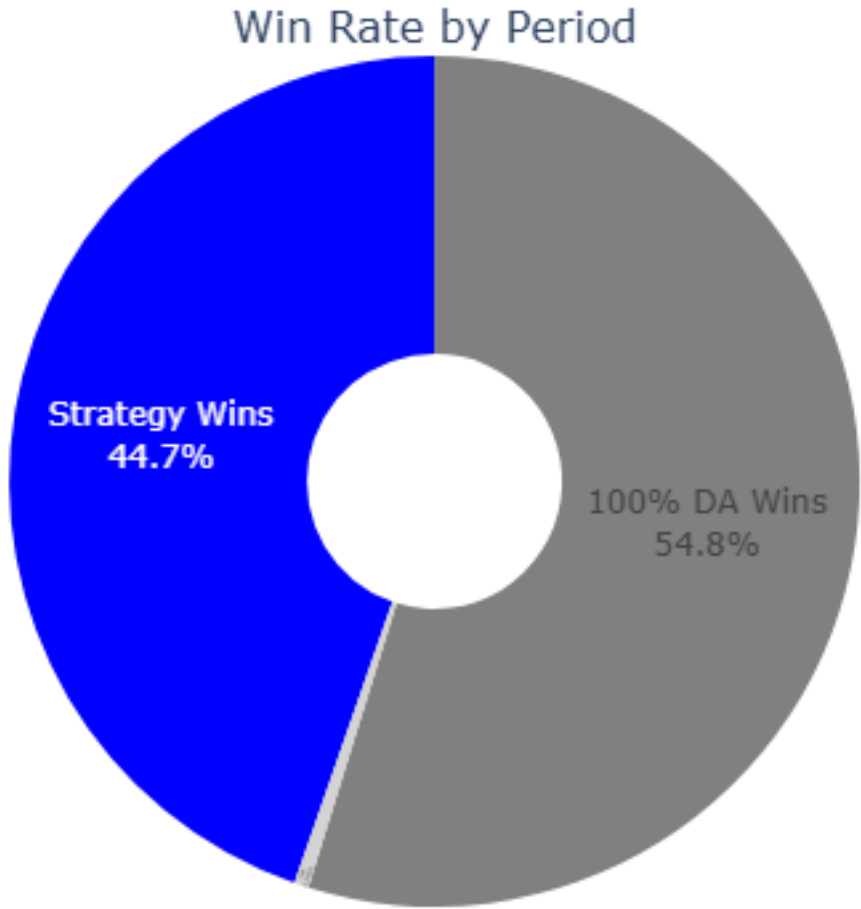


Split Analysis

80% DA + Remaining % IDA1

- A comparison of each individual traded period can tell us whether the DA strategy won through consistent smaller wins or occasional large wins.
- 100% DA had a better win rate.
- 100% DA had a greater win total.
- 80/20% IDA1 had a greater average win.
- 80/20% IDA1 also had a greater max win.
- This suggests that the 80/20% split is more volatile but is not as consistent as 100% DA.
- Overall trading 100% DA was better.

Big Wins vs Consistency Analysis (2025-04-01 to 2026-03-31)
Strategy wins 163/365 periods | 100% DA wins 200/365 periods



Metric	100% DA	80/20%
Win Rate	54.8%	44.7%
Total Win	€1,443,306	€1,312,130
Average Win	€7,217	€8,050
Maximum Win	€86,020	€94,601

Split Analysis

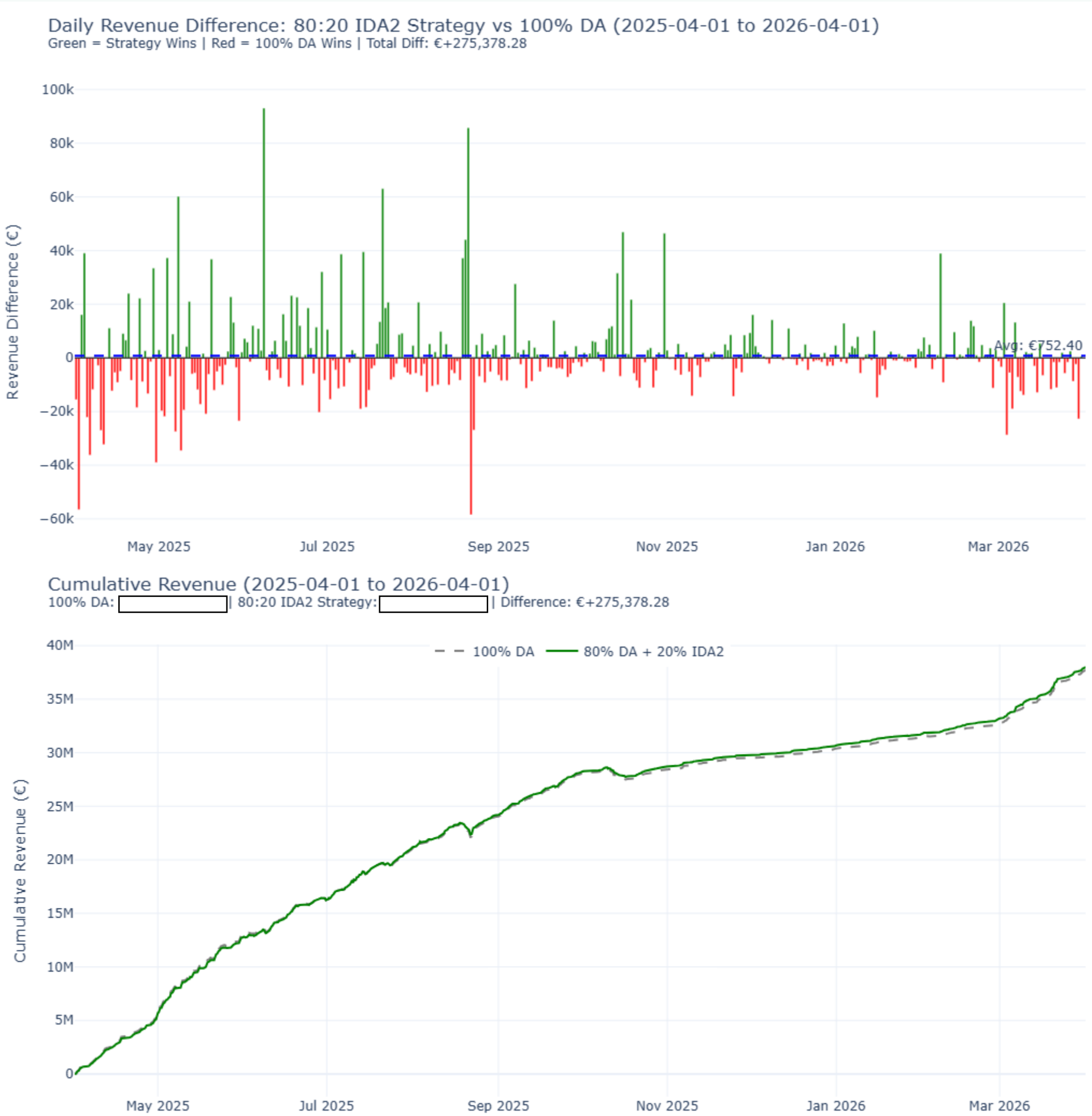
80% DA + Remaining % IDA2

- The same test was done with IDA2 instead of IDA1.
 - 80% of the volume was traded to the DA forecast, approx. 20% to the IDA2 forecast, and the remainder was settled in the imbalance auction.
 - This was done to investigate whether we could capture more value in other auctions.

- Strategy results over a year:

Strategy	Value
80% DA + ~20% IDA2 Revenue	€XXX
100% DA Revenue	€XXX
Difference	+ €275,378.28

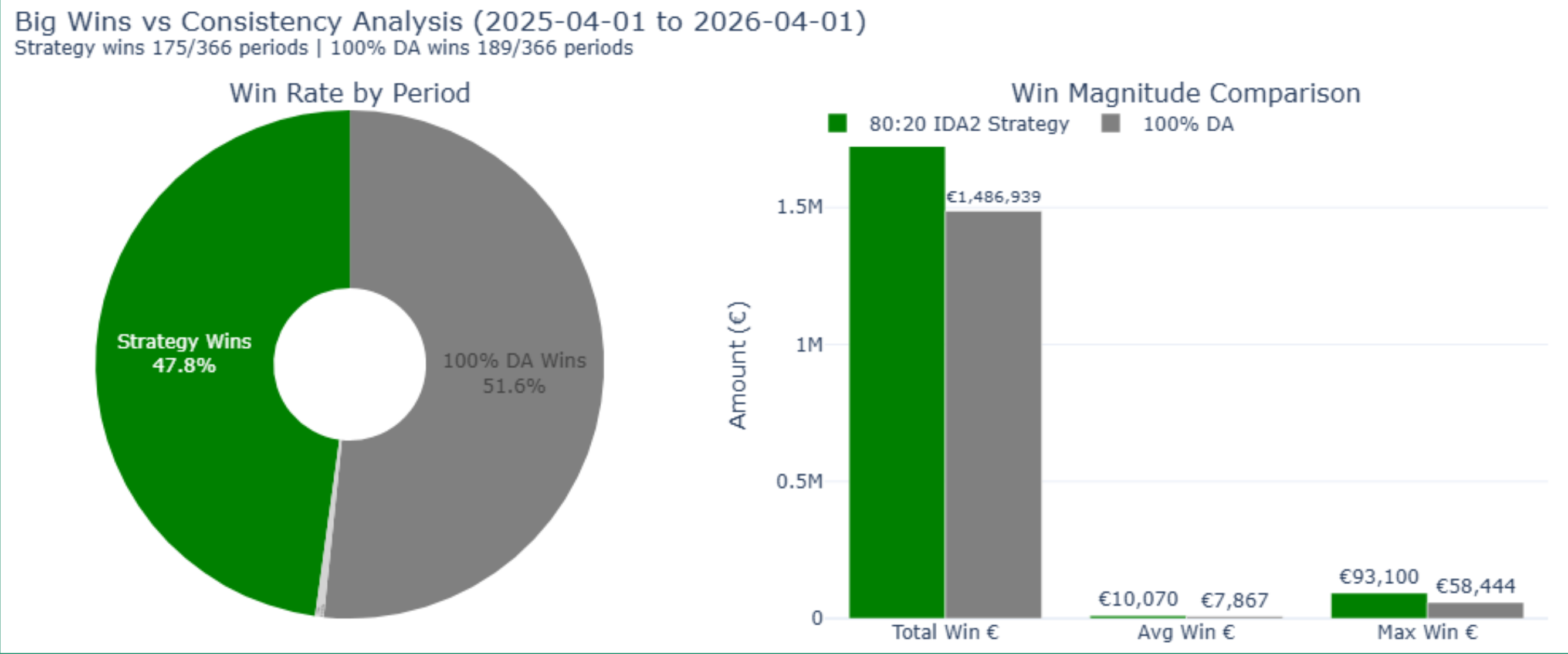
- The IDA2 split outperformed trading 100% to DA.



Split Analysis

80% DA + Remaining % IDA2

- 100% DA had a better win rate.
- 80/20% IDA2 had a greater win total.
- 80/20% IDA2 had a greater average win.
- 80/20% IDA2 also had a greater max win.
- This suggests that the 80/20% split is more volatile but is not as consistent as 100% DA.
- The IDA2 split performed better than 100% DA.
- The IDA2 split outperformed the IDA1 split.



Metric	80/20% IDA1	80/20% IDA2
Total Revenue	€XXX	€XXX
Difference		+ €443,444.43

Metric	100% DA	80/20%
Win Rate	51.6%	47.8%
Total Win	€1,486,939	€1,762,317
Average Win	€7,867	€10,070
Maximum Win	€58,444	€93,100

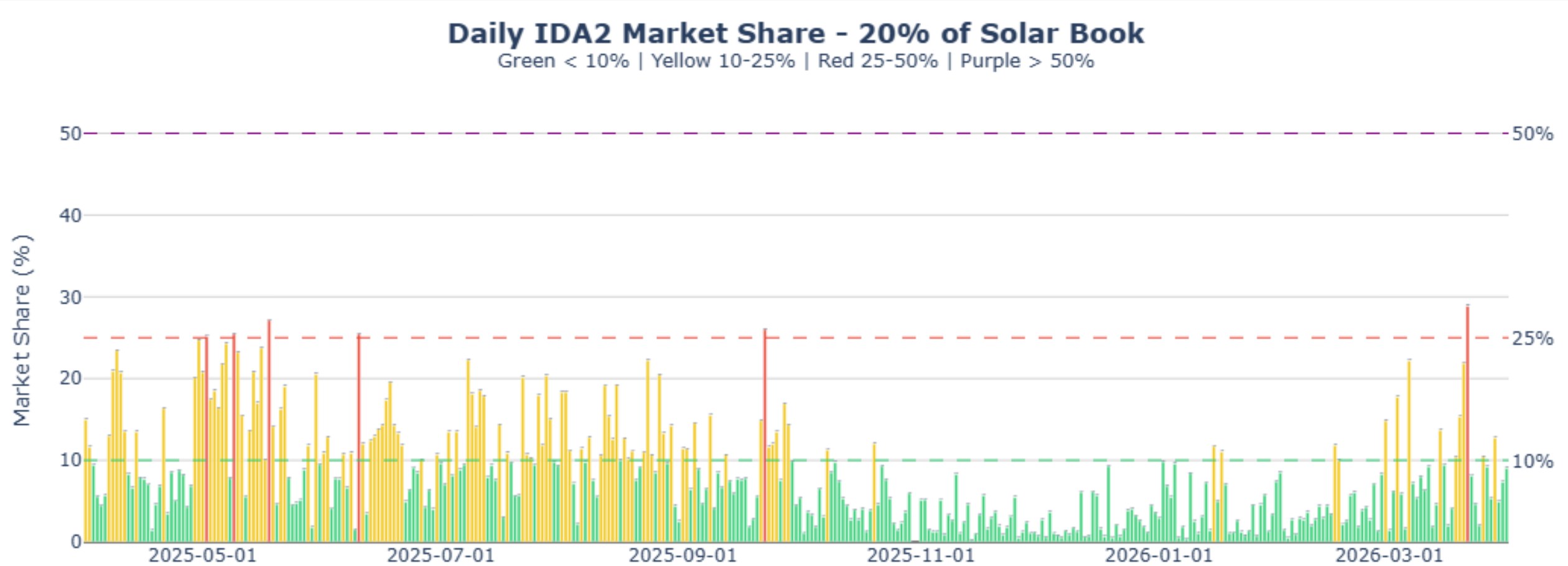
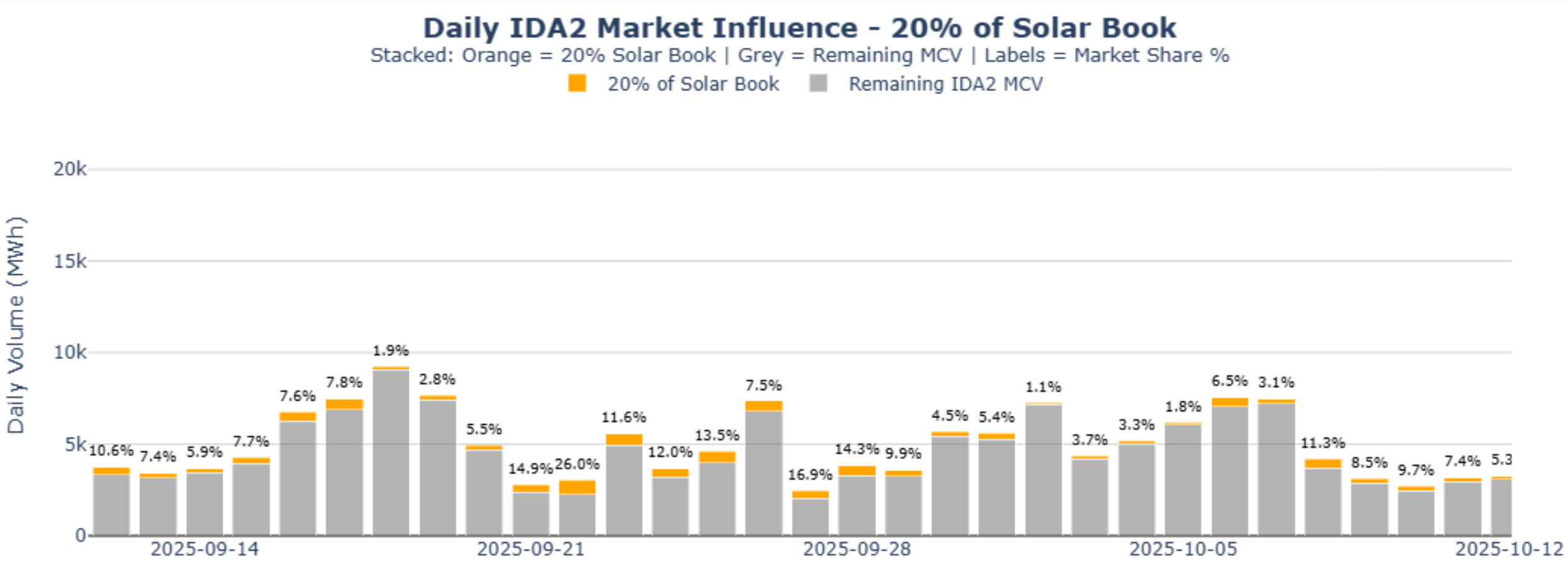
Our Influence on IDA2 MCV

- If we had traded 20% of this volume at IDA2, we would have had the following influence on the market:

Overall Statistics	
Solar Book (Total)	750,246 MWh
20% of Solar Book	150,049 MWh
IDA2 MCV (Total)	2,217,451 MWh
Average Market Share	9.73%
Max Market Share	90.66%

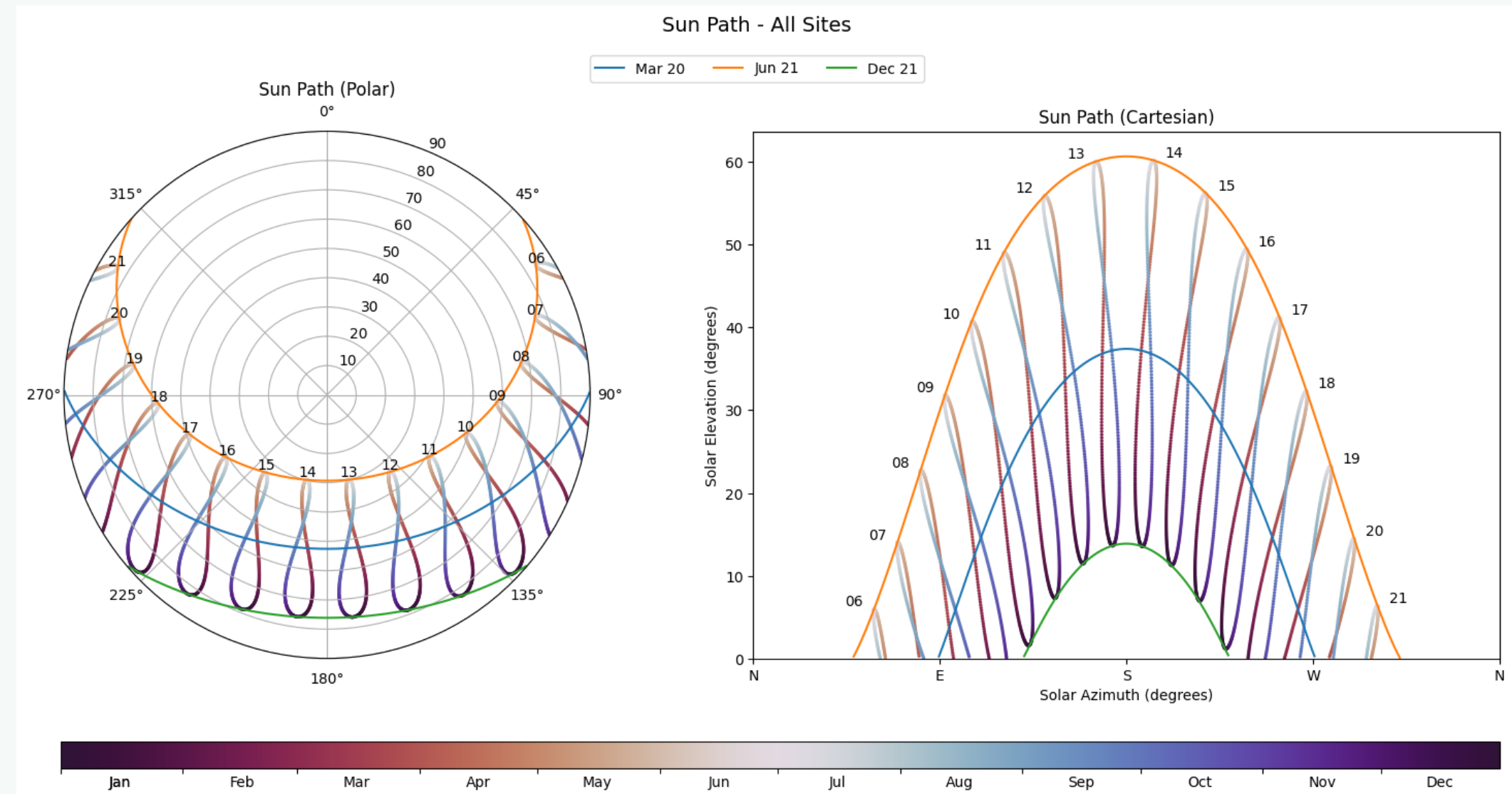
30-min Periodic Analysis	
Periods with > 10% Share	2,539 (26.9%)
Periods with > 25% Share	567 (6.0%)
Periods with > 50% Share	50 (0.5%)

Daily Analysis	
Days with > 10% Share	112 (30.9%)
Days with > 25% Share	6 (1.7%)
Days with > 50% Share	0 (0.0%)



80-100% DA + IDA1

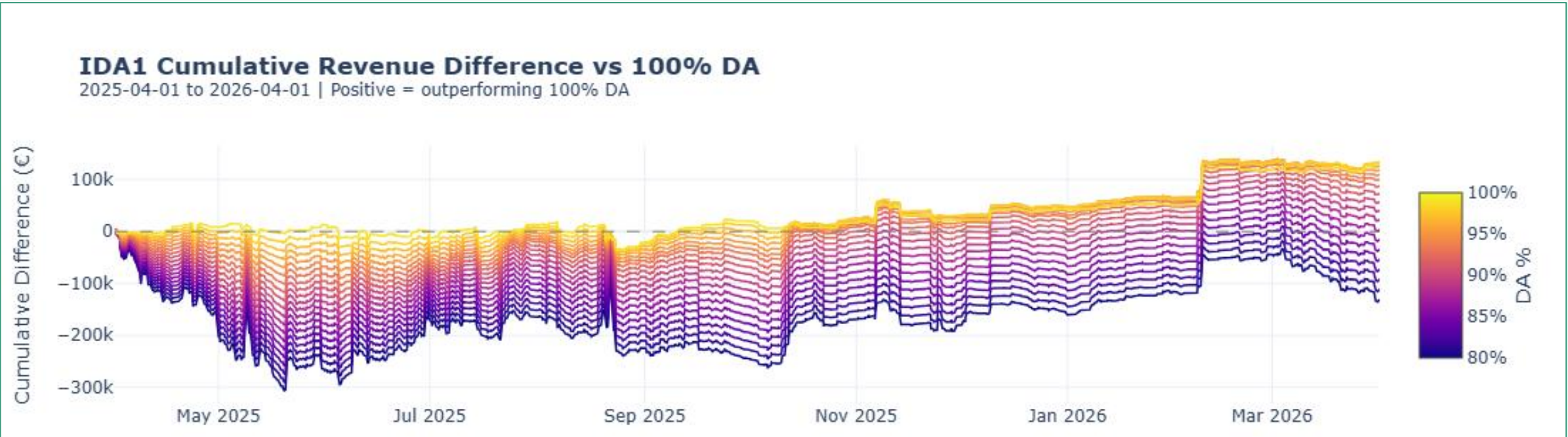
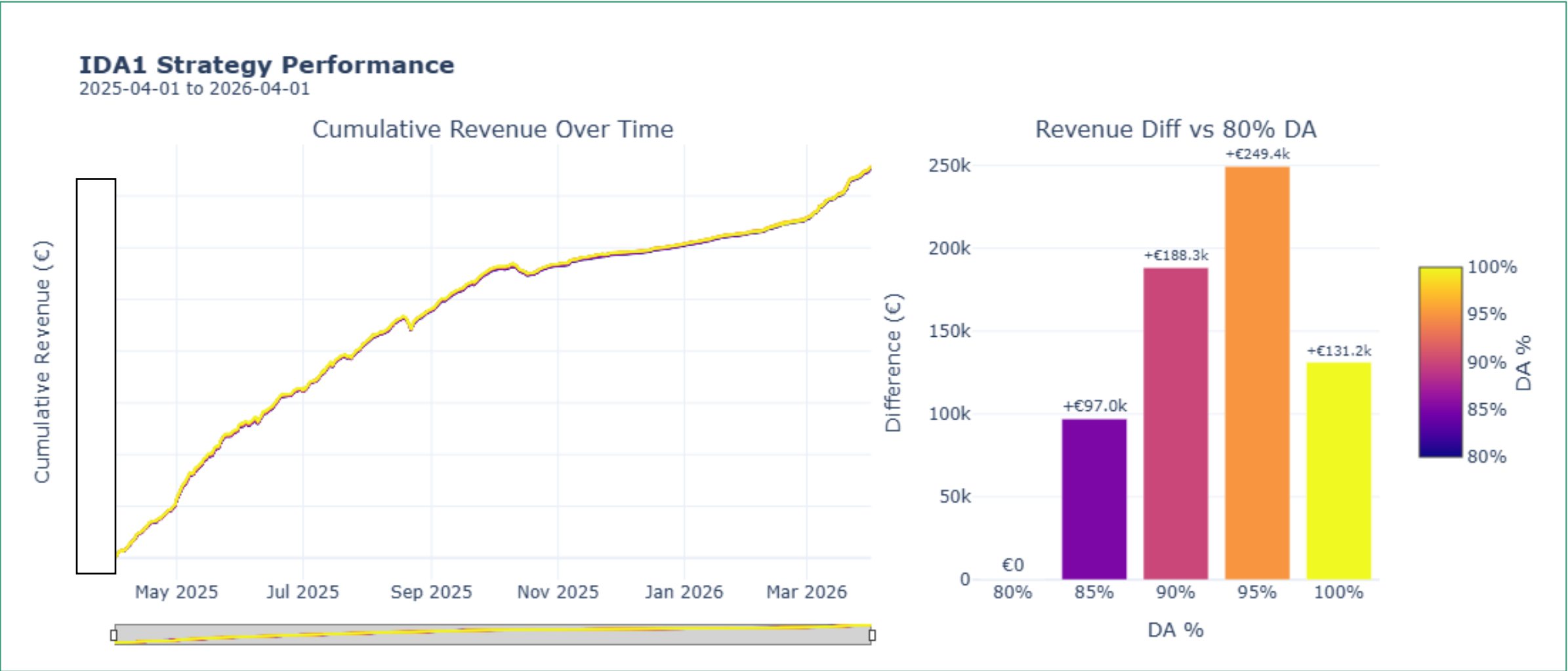
- Despite running splits for 80% DA + remaining volume in IDA1/IDA2 and 100% DA, the optimal split may lie somewhere in the range of 80 to 100.
- In order to test this, splits were run in increments of 1%, starting at 80% DA and ending at 100% DA.
- The results of each split were plotted alongside each other, which allows us to observe their changes in performance across a year.
- Factors such as high DA, IDA1/2 or imbalance prices driven by market events can cause the strategies to change direction or gain/lose momentum, which is reflected in the plots.



80-100% DA + IDA1

- Splitting over DA and IDA1 produced fairly consistent results. Across the year, lower splits (purple) performed worse.
- The higher splits consistently performed better with the upper-middle splits generating the most revenue from November 2025 onward.
- Overall, the best split was 98% DA.
- The results suggest that there is value in holding back some volume to trade in IDA1.

Metric	Strategy	Revenue
Best IDA1 Strategy	98% DA	€XXX
Baseline Strategy	100% DA	€XXX
Difference		€133,575.61



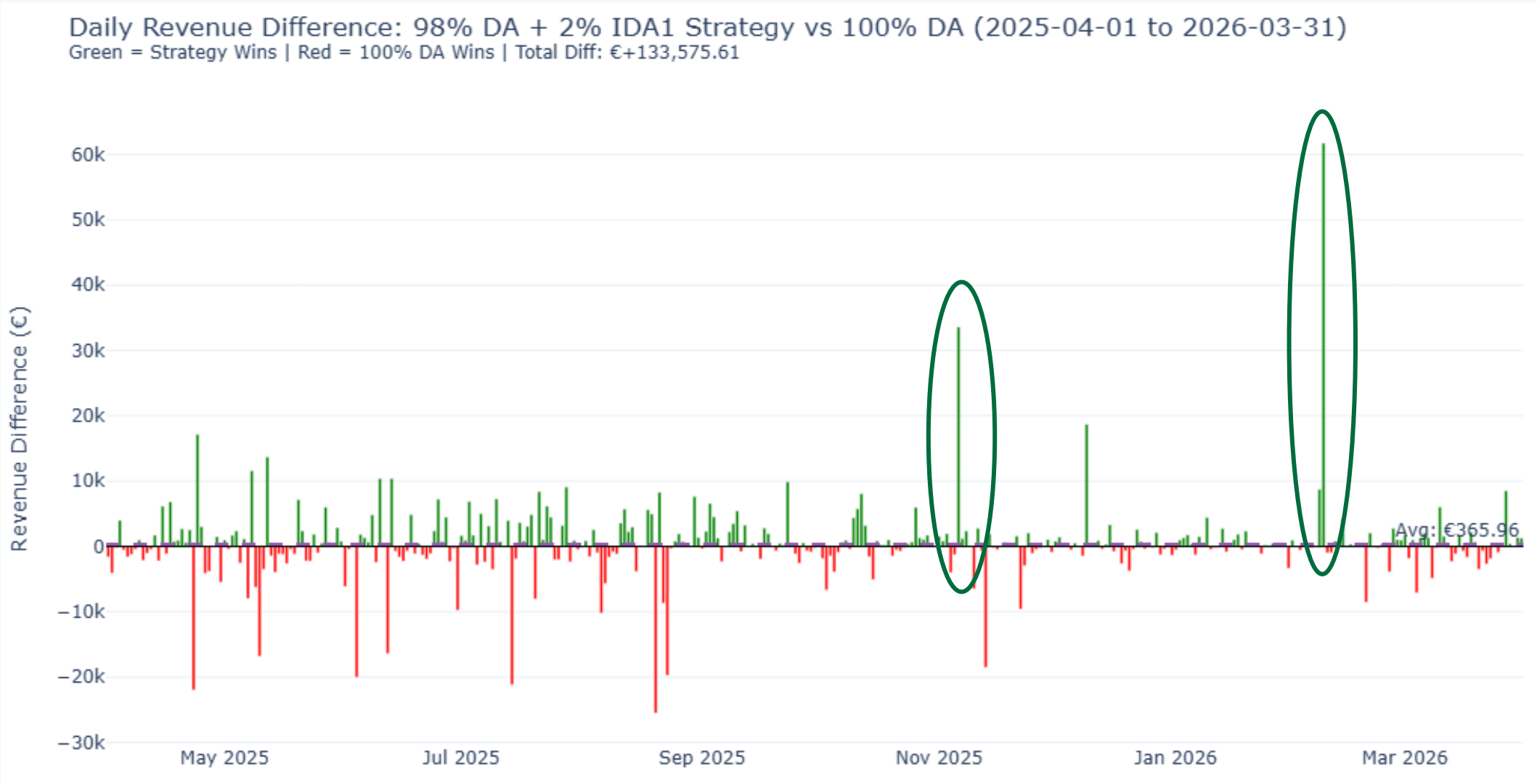
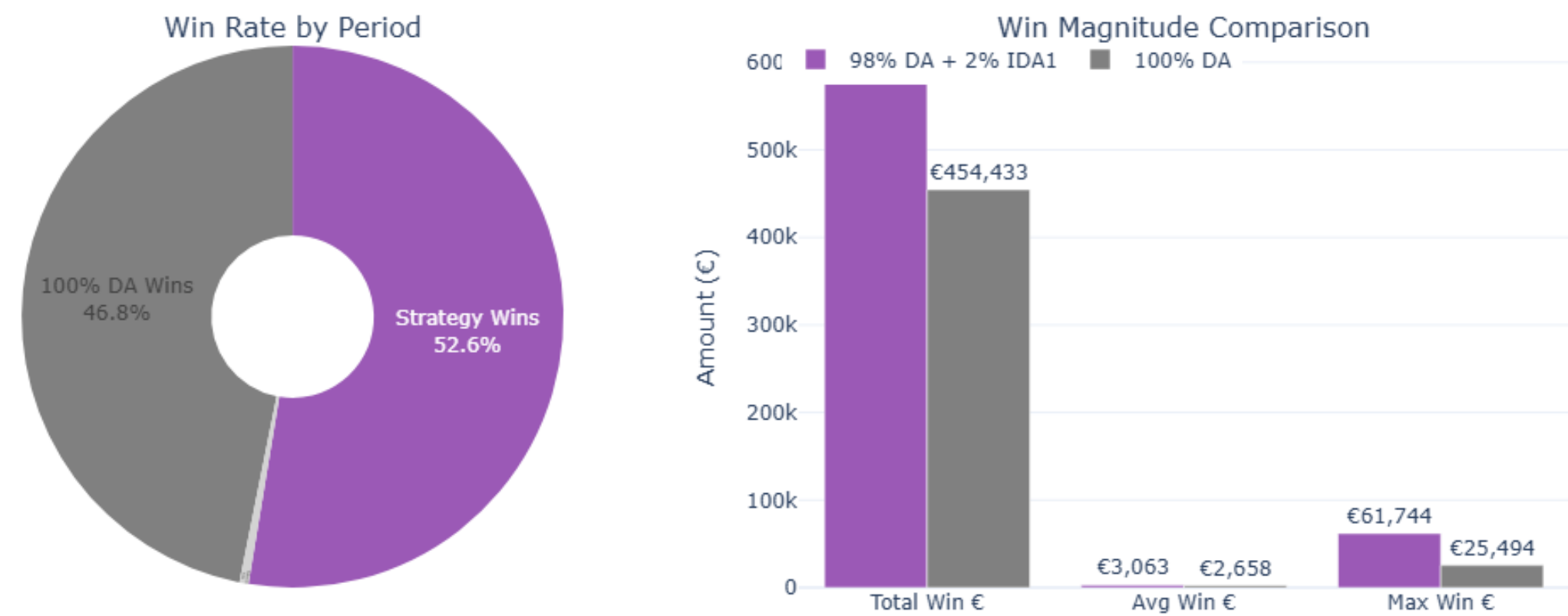
Optimal IDA1 Split

98% DA + Remaining % IDA1

- The strategy won 52.6% of the time compared to 100% DA.
- It also had a greater total win, average win, and maximum win.
- However it did have two days of massive wins, outlined in the figure to the right, that could have driven the result.
- But overall it was more consistent too.

Metric	100% DA	98/2%
Win Rate	46.8%	52.6%
Total Win	€454,433	€588,008
Average Win	€2,658	€3,063
Maximum Win	€25,494	€61,744

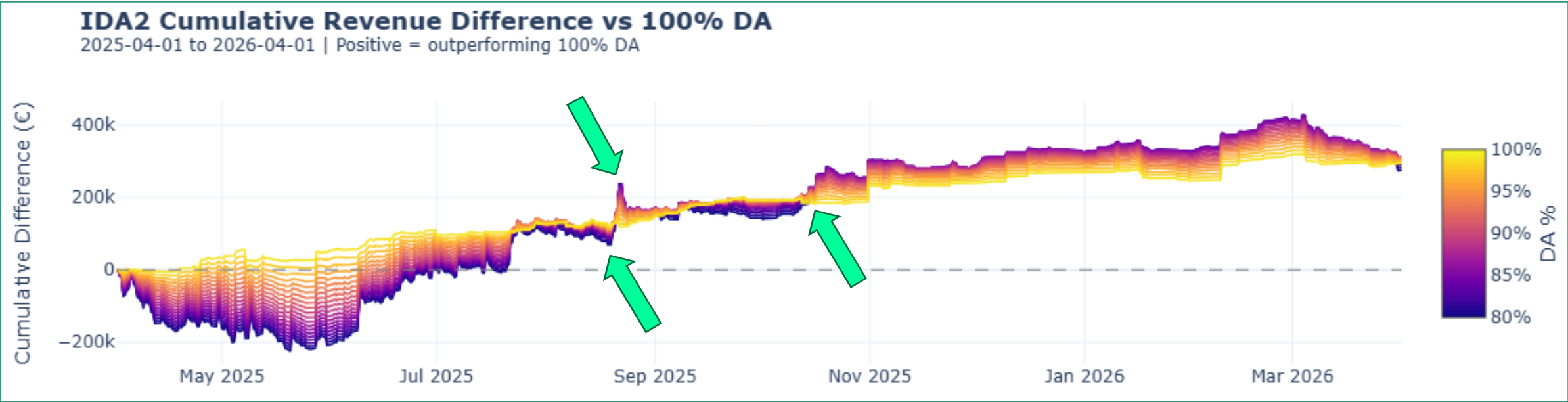
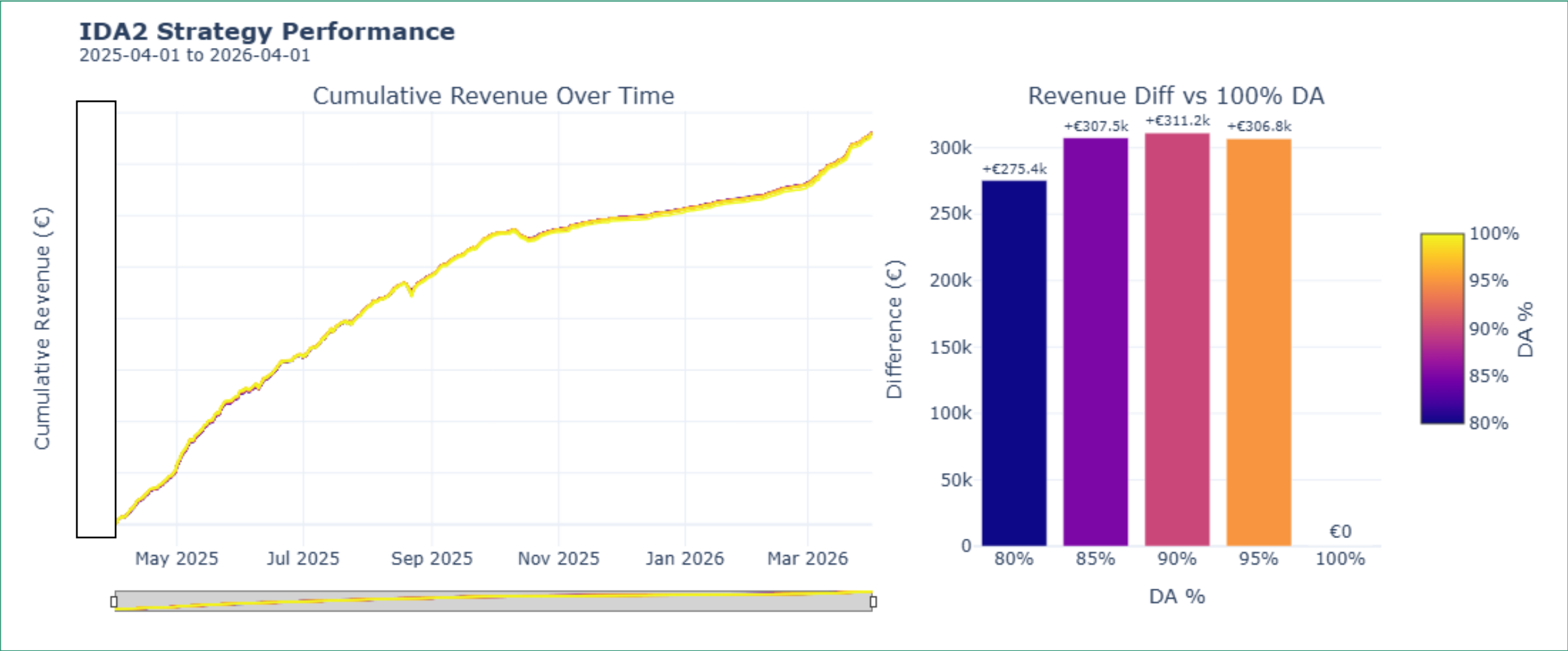
Big Wins vs Consistency Analysis (2025-04-01 to 2026-03-31)
Strategy wins 192/365 periods | 100% DA wins 171/365 periods



80-100% DA + IDA2

- The expected price volatility of IDA2 can be seen here, especially on August 21st and 22nd where high forecast error caused significant gains and losses.
- This volatility also resulted in multiple inflection points to be generated along the curve, highlighted by the arrows below.
- Overall, the IDA2 strategy outperformed the IDA1 strategy, with both the best and worst IDA2 strats outperforming the best and worst IDA1 strats.
- However, the IDA2 strategy yielded greater dispersion, which once again reflects this higher volatility and greater risk.

Metric	Strategy	Revenue
Best IDA2 Strategy	90% DA	€XXX
Baseline Strategy	100% DA	€XXX
Difference		€311,207.81

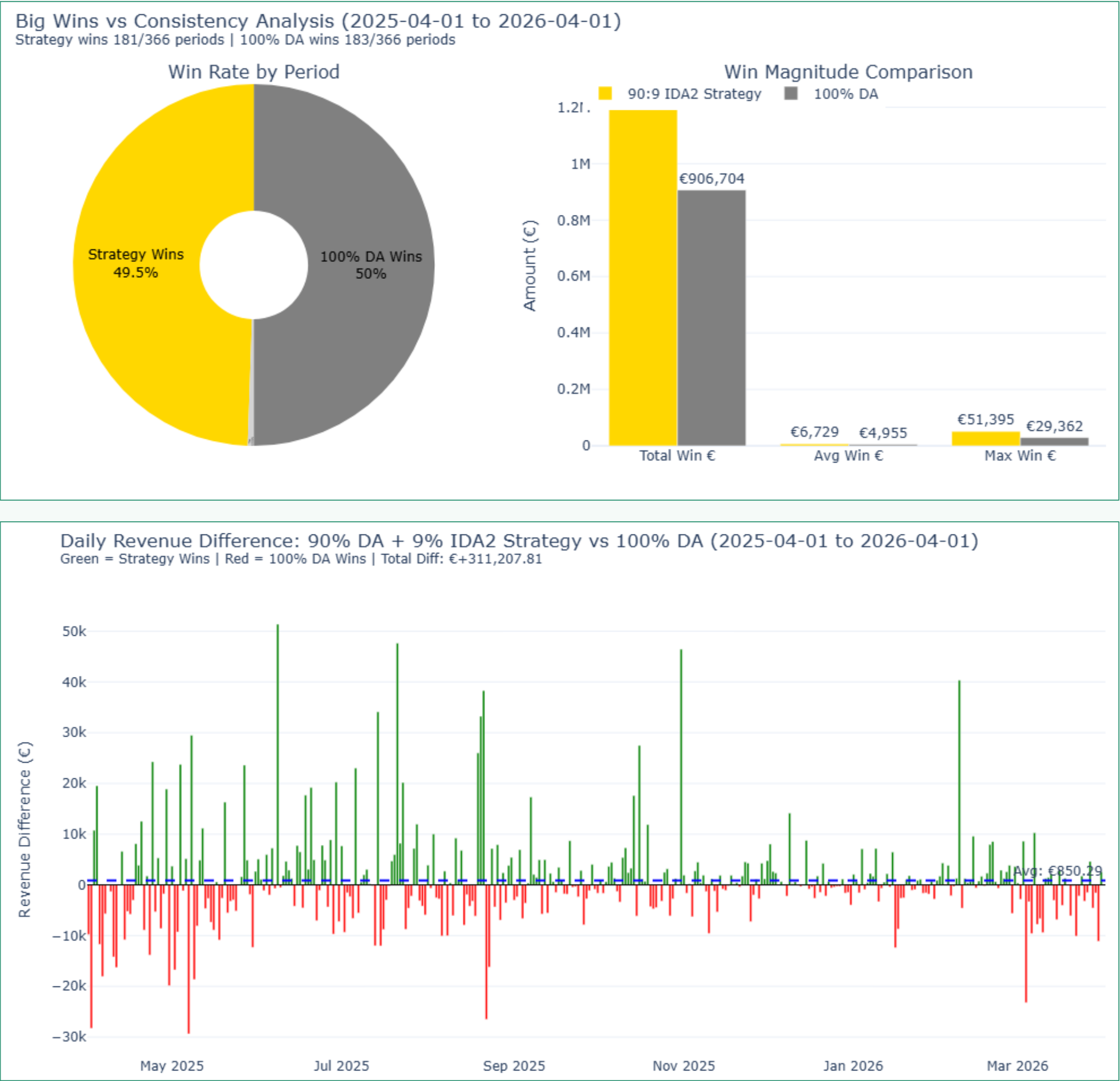


Optimal IDA2 Split

90% DA + Remaining % IDA2

- In terms of consistency, the 100% DA split outperformed the 90% split slightly.
- The 90% split had a greater total win, average win, and maximum win.
- From the graph below it looks like there were multiple high magnitude revenue days driving the performance.

Metric	100% DA	98/2%
Win Rate	50%	49.5%
Total Win	€906,704	€1,217,911
Average Win	€4,955	€6,729
Maximum Win	€29,362	€51,395



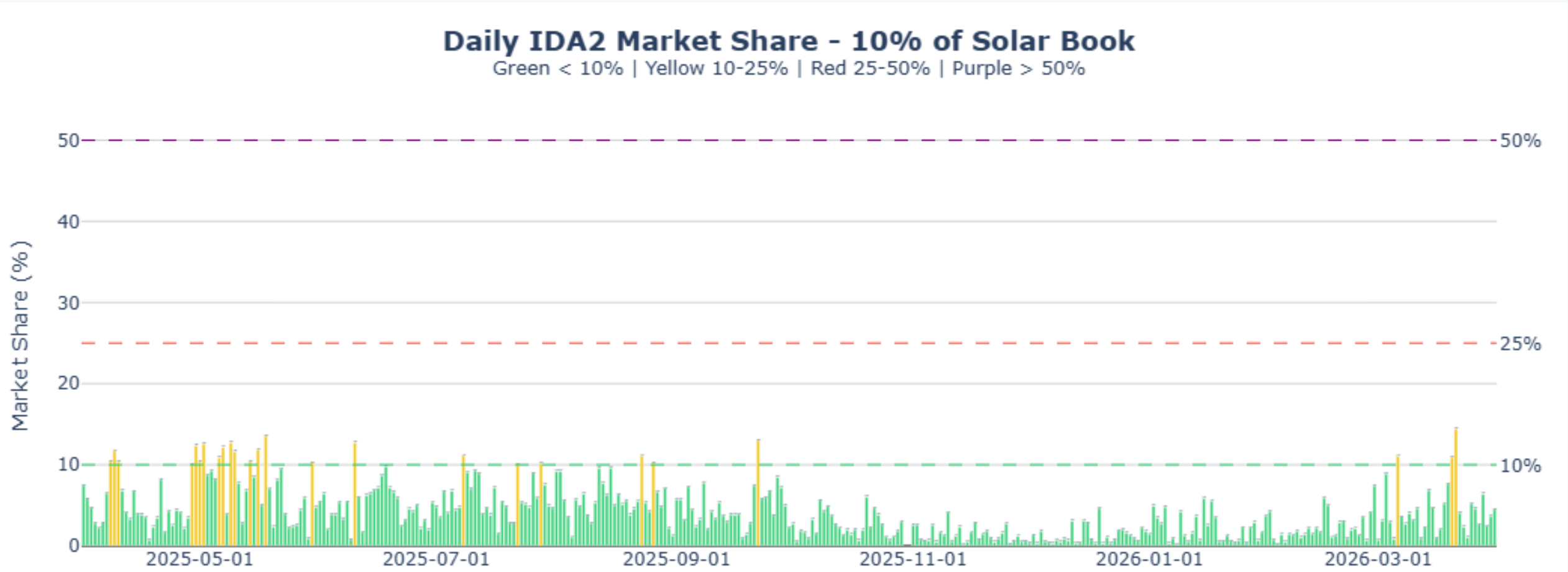
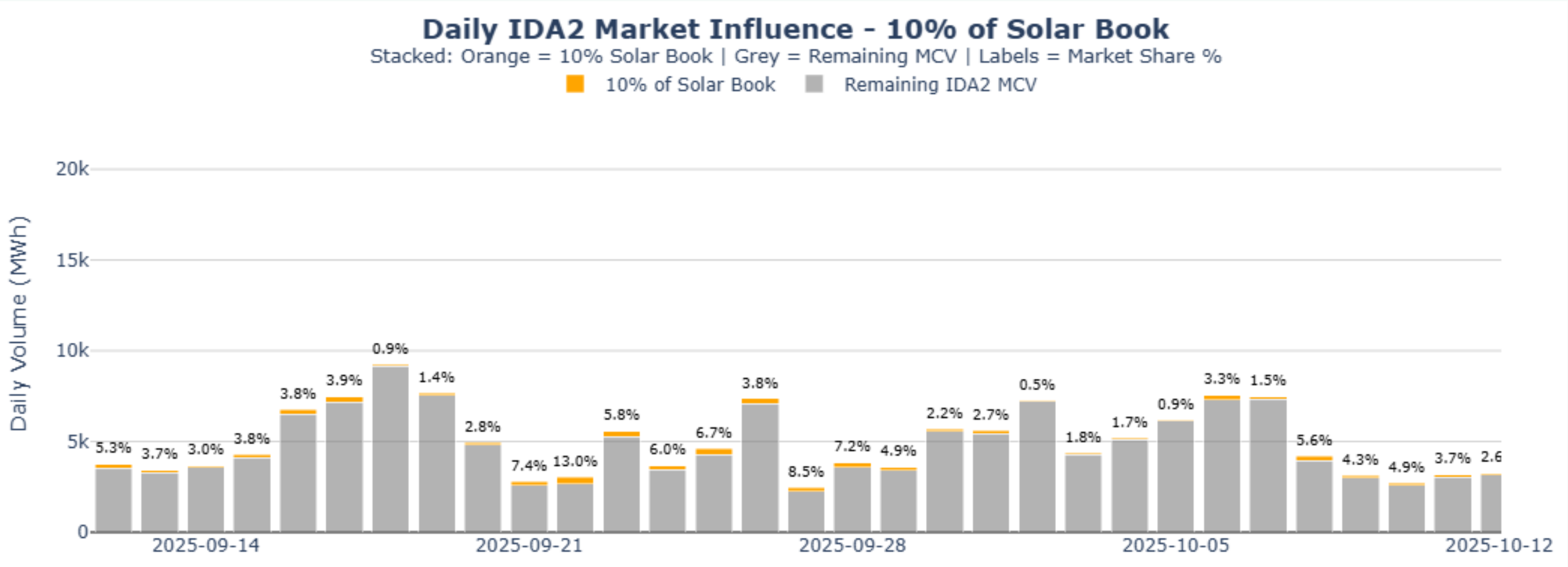
Our Influence on IDA2 MCV

- If we had traded 10% of this volume at IDA2, we would have had the following influence on the market:

Overall Statistics	
Solar Book (Total)	750,246 MWh
20% of Solar Book	75,025 MWh
IDA2 MCV (Total)	2,217,451 MWh
Average Market Share	4.86%
Max Market Share	45.33%

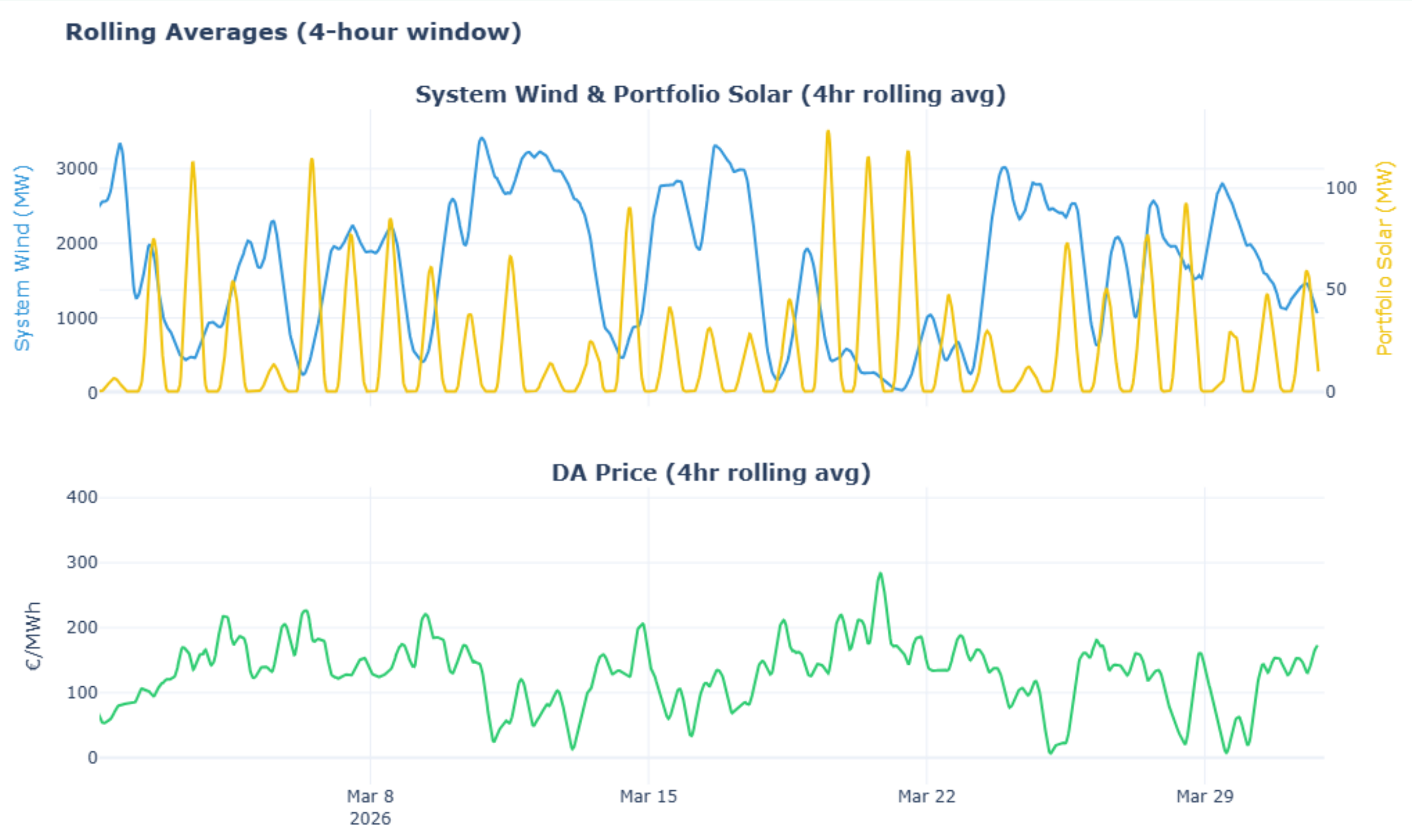
30-min Periodic Analysis	
Periods with > 10% Share	949 (10.1%)
Periods with > 25% Share	50 (0.5%)
Periods with > 50% Share	0 (0.0%)

Daily Analysis	
Days with > 10% Share	25 (6.9%)
Days with > 25% Share	0 (0.0%)
Days with > 50% Share	0 (0.0%)



Influence of Wind on Prices

- We know that high wind tends to push prices down, but here are some stats that back this up.



Key Statistics:

Wind vs DA Price Correlation

- 0.454

- Moderate negative relationship. As wind increases, DA price tends to decrease.

Wind-Price Sensitivity

- 0.025 €/MW

- For every 1 MW increase in system wind, the DA price drops by €0.025 (-€25.08 per 1000 MW).

R-Squared (R^2)

0.206

- Roughly 21% of DA price variance is driven by wind alone. Other factors such as demand, solar, gas prices etc. drive the remaining 79%.

P-Value (P)

~ 0.00

- The relationship is statistically significant and is not based on random chance.

Influence of Wind

Example: August 2025

- Here is plot for August 2025, a particularly high solar month where we had large balancing costs.
- Wind levels were low, solar levels were high, and prices remained relatively stable across DA, IDA1 & IDA2. However the imbalance prices were very volatile.
- When wind came up during these periods prices were pushed down, with imbalance prices being completely floored in some instances.
- The opposite is also true, low wind and high solar was extremely volatile in the BM.
- It is clear that seasonality plays a significant role in determining how effectively we can trade solar (i.e. large imbalances in the summer have greater impact).
- Further analysis is required to investigate why solar is being undersold on low wind days — specifically whether this is driven by wind forecast error or an expectedly low wind output.
- Based on percentages of the maximum, wind levels were categorised into 'High' (>66%), 'Medium' (33-66%), and 'Low' (<33%).

FY25/26

Wind	Periods	Wind (MW)	DA (€)	IDA1 (€)	IDA2 (€)	Imbalance (€)	DA vs Low Wind
Low	2940	344	140.1	138.95	138.36	152.11	-
Medium	2832	1150	104.5	101.94	98.91	100.39	-35.59
High	2581	2365	85.28	84.19	80.29	78.38	-54.81



Aug 25

Wind	Periods	Wind (MW)	DA (€)	IDA1 (€)	IDA2 (€)	Imbalance (€)	DA vs Low Wind
Low	256	181	135.39	128.48	134.99	181.99	-
Medium	308	792	96.63	96.82	91.31	109.51	-38.76
High	269	1,908	69.32	69.00	66.10	55.94	-66.08

Curtailment Analysis

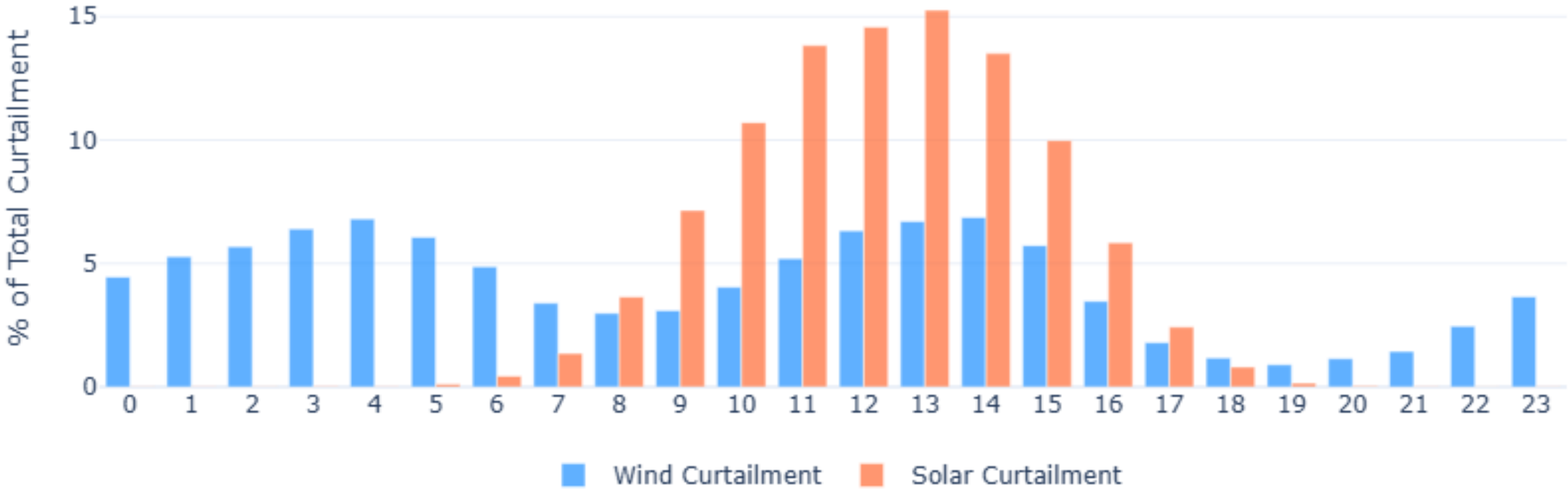
- It is worth investigating whether there is any correlation between solar and wind curtailment.

Category	Count	Percentage
Total 30-min Periods	17,518	100%
Neither Curtailed	13,196	77.1%
Wind Only Curtailed	1,881	10.7%
Solar Only Curtailed	163	0.9%
Both Curtailed	1,963	11.2%

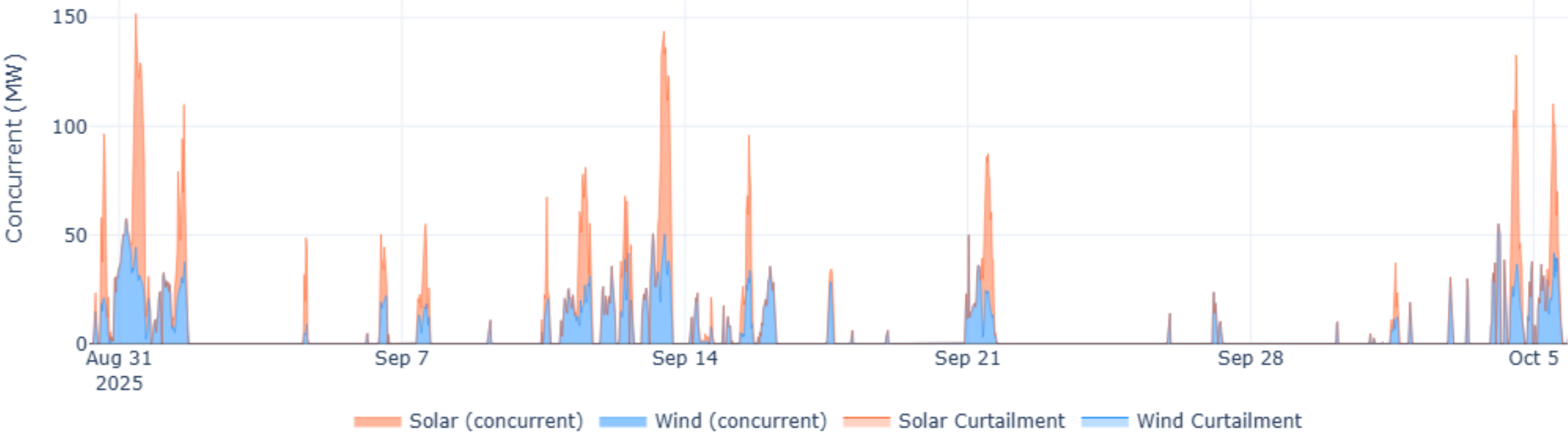
Metric	Value (MW)	Percentage
Total Wind Curtailment	68,386.4	65.4% of total
Total Solar Curtailment	36,181.2	34.6% of total
Wind During Concurrent	40,583.0	59.3% of wind
Solar During Concurrent	35,617.0	98.4% of solar

- Based on this analysis, it is fair to say that there is a relationship between solar and wind curtailment – and therefore forecasting wind curtailment could possibly help us trade solar more effectively.
- Further analysis could include studying the impact of over selling solar on high wind days.

Curtailment Distribution by Hour of Day
% of total curtailment occurring in each hour



Concurrent Curtailment (Both Wind & Solar Curtailed)

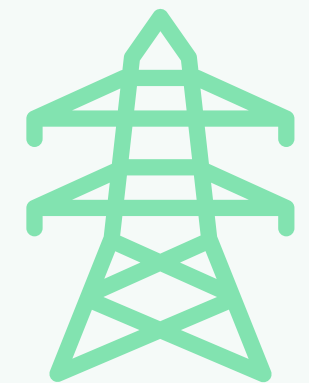
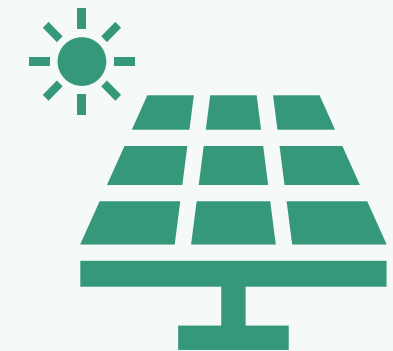


Conclusions

Based on this analysis, we can conclude that there are multiple different factors that direct and indirect influences on our solar book performance.

- **Forecast Accuracy** is without doubt the most influential factor on our performance. From this analysis, we have noticed a systematic over-forecasting bias – which consequently has caused us to be short.
- **Curtailment** creates significant unavoidable exposure particularly during peak solar hours.
 - Further analysis needs to be done on the individual sites such as [SITE NAME] and [SITE NAME] to determine why the magnitude of over-forecasting and error are so large.
- **Wind** is the key driver of Irish electricity prices. High wind periods correlate with both lower prices and increased solar curtailment risk, making wind forecasts essential for optimising our solar trading strategy.
- Depending on which auction we trade our volume in, we can have a significant influence on the **Market Clearing Volume**. This means that our trading activity has the potential to influence prices and imbalance volumes – which is something we need to be careful of.
 - A practical example of this was simulating 20% of our solar volume through IDA2, representing 10% of the total MCV.

Addressing forecast bias and understanding site-level performance variations represent the highest-impact opportunities for improving our solar book profitability.



Next Steps

Strategy Development and Testing

- Strategy Development Aim – optimise volume in the IDA’s to reduce balancing exposure.
- How - To optimise the volume between auctions, proposing to use a scoring system to categorise the forecast on a given day.
- The system looks to categorise the forecast based on volume levels, peak level and ramp rate. Seasonality is reduced using NSI (normalised solar index Forecast/14day rolling median - in winter you can extend the formula to include seasonal average using 2/3 years of data).
- Backtest the scores with associated % split to see the most profitable (if that is true at all)
- If it’s true, shadow trade.
- Investigate underselling solar on low wind days and adding as a screen to the scoring

Date	Month	Forecast MWh	14d Median Expected	NSI	NSI Bucket	Peak MW	Peak	Peak Bucket	Max Ramp MW	Ramp	Ramp Bucket	Score	DA Sell %	IDA 2 Sell %
01-Apr	Apr	95	118	0.81	Medium	19	0.63	Medium	4	0.13	Low	4	95.0%	5.0%
02-Apr	Apr	140	120	1.17	High	27	0.90	Medium	8	0.27	Medium	4	95.0%	5.0%
03-Apr	Apr	155	122	1.27	High	29	0.97	High	6	0.20	Medium	5	97.5%	2.5%
04-Apr	Apr	50	121	0.41	Low	15	0.50	Low	5	0.17	Medium	1	85.0%	15.0%
05-Apr	Apr	180	125	1.44	High	30	1.00	High	9	0.30	High	4	95.0%	5.0%
06-Apr	Apr	125	128	0.98	Medium	24	0.80	Medium	5	0.17	Medium	3	92.5%	7.5%

NSI	NSI = Forecast Daily MWh / 14-day rolling median expected MWh.
Peak	Peak = Forecast Peak MW / Installed Capacity. For a 30 MW site, Peak = 1.00 means the forecast reaches full output.
Ramp	Ramp = Max half-hour MW change / Installed Capacity. For a 30 MW site, Ramp = 0.33 means the largest half-hour forecast step is 10 MW.
Scoring	This example uses a simple additive score: NSI Low/Medium/High = 0/1/2; Peak Low/Medium/High = 0/1/2; RampAdj Low/Medium/High = 2/1/0. Lower ramp gets a higher score because smoother days are typically easier forecast

Normalised Solar Index (NSI) - Example

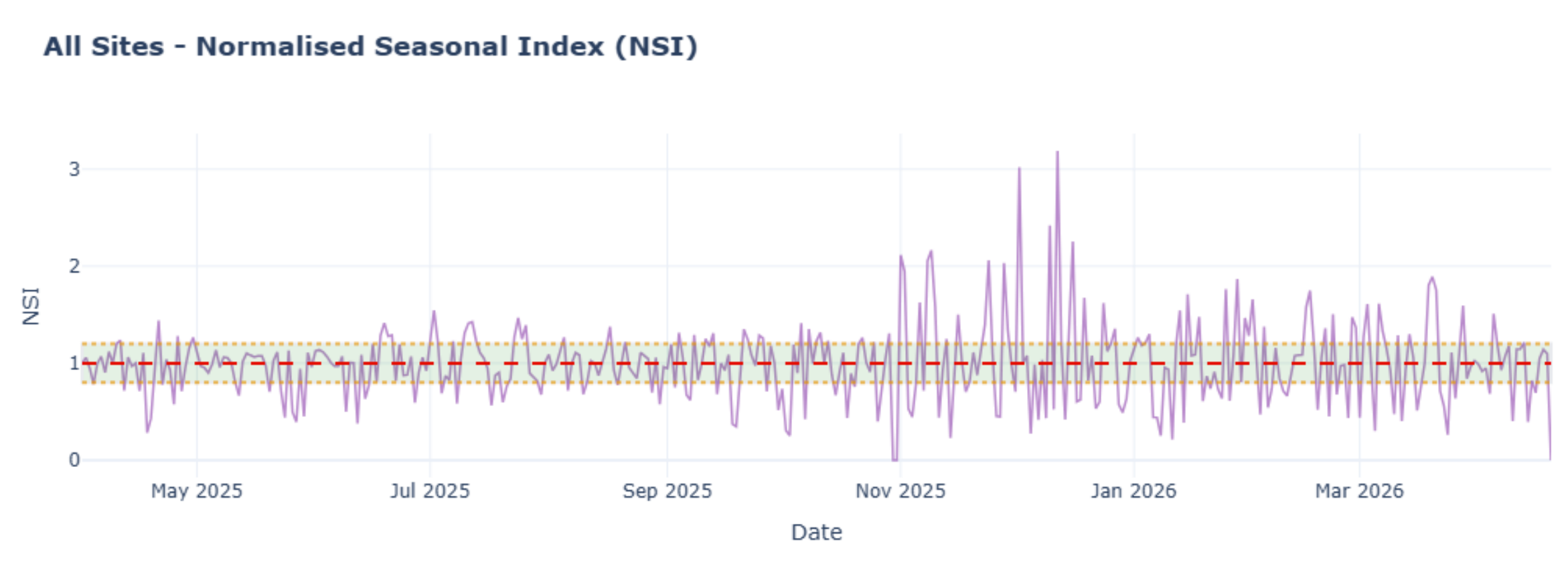
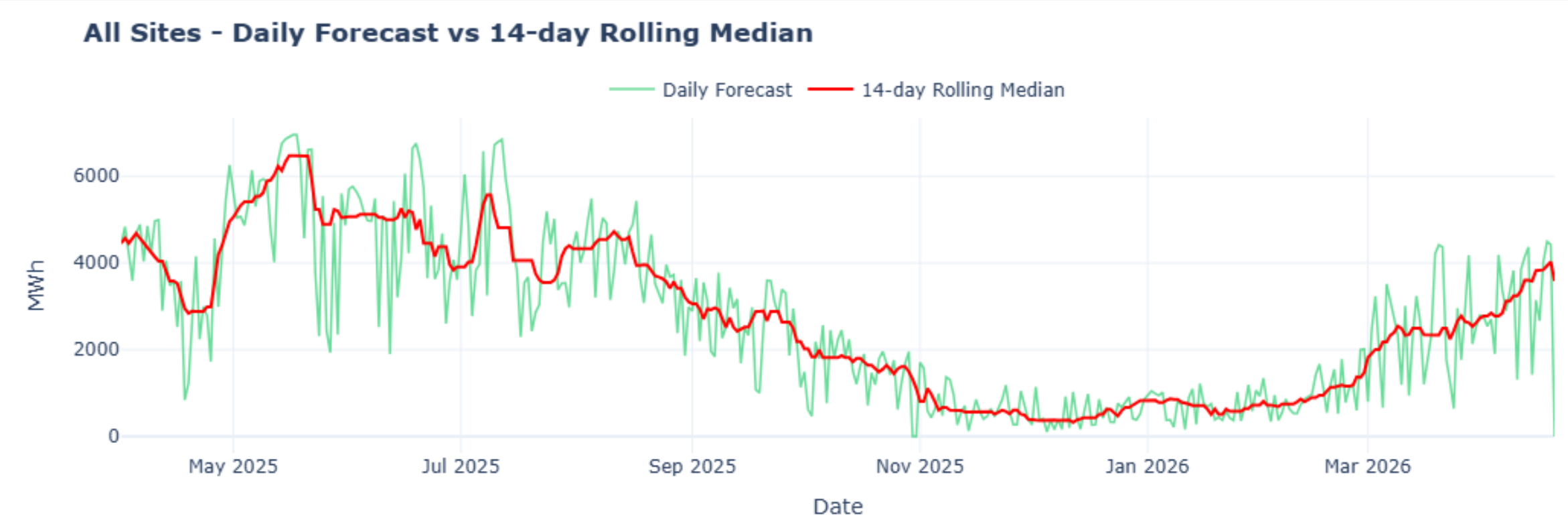
- A rolling median can also help us determine the seasonality of solar generation by creating a smoother curve.
- However this doesn't necessarily give us a good edge to predict the solar output due to its naturally high daily volatility – particularly in the summer months as seen on the right.
- A normalised solar index was defined as:

$$NSI = \frac{\text{Daily Total Forecasted Volume}}{14 - \text{Day Rolling Median}}$$

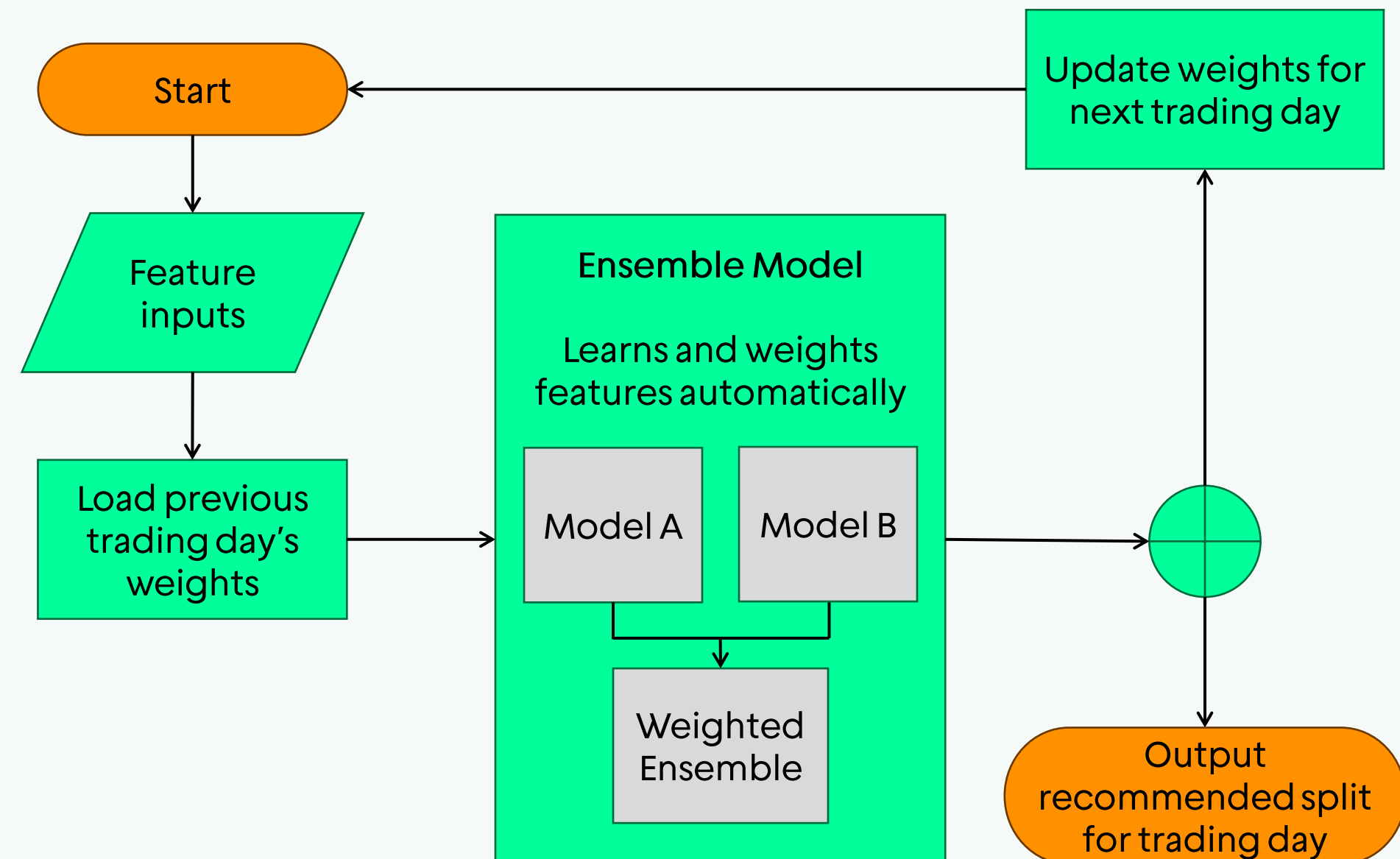
NSI Value	Meaning
NSI = 1.0	Exactly average day (matches rolling median)
NSI > 1.0	Above-average solar day (better than recent)
NSI < 1.0	Below-average solar day (worse than recent)
NSI > 1.2	Exceptionally good day (20% above recent)
NSI < 0.8	Exceptionally poor day (20% below recent)

- Results:

Metric	Value
Average NSI	0.997
NSI Standard Deviation	0.402
Days Above Normal (> 1.2)	93 (24.2%)
Days Below Normal (< 0.8)	107 (27.8%)
Days Normal (0.8 – 1.2)	185 (48.1%)



How Could This Be Done?



Example Feature Inputs	Weights
Solar Generation Forecasts (DA, IDA1, IDA2)	...
Wind Generation Forecasts (DA, IDA1, IDA2)	...
System Demand Forecast	...
Historical Forecast Error Metrics (ME, MAE, RMSE)	...
Forecast Bias History (systematic over/under-forecasting)	...
Auction Price History (DA, IDA1, IDA2, Imbalance)	...
Price Spreads (DA-IDA1, IDA1-IDA2, IDA2-Imbalance)	...
Historical Price Volatility by Auction	...
Imbalance Price Bias (long vs short penalty)	...
SNSP Forecast	...
Curtailment Events	...
Historical MCV %	...
Hour of Day & Day Type (Weekday/Weekend)	...
Season / Time of Year	...
Expected Ramp Rate (MW/hour)	...
Sum	1.00

- This is a revenue maximisation problem that can be solved by predicting what the optimal auction split for a given day will be.
- By using some of the data features we have just analysed, alongside our forecasts, a recommendation can be made for the following trading day on what the best volume split would be to maximise revenue.
- This could be done by using an ensemble or some form of regression, where each feature is an input to the model and can be dynamically weighted from 0-1, depending on how much influence it would have on a particular day.



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